



A Survey of Data Augmentation in Domain Generalization

Yingyi Zhong¹ · Wen'an Zhou¹ · Zhixian Wang¹

Accepted: 24 February 2025 / Published online: 22 March 2025
© The Author(s) 2025

Abstract

Most machine learning algorithms typically assume that the data distribution of the training and test sets are consistent, but this assumption often fails to hold in practical applications. Domain generalization aims to train a model using only available source data so that the model can generalize to unseen domains. Data augmentation is an important technique in domain generalization, but there are few comprehensive reviews investigating and summarizing its use in domain generalization. This study provides a comprehensive literature review of data augmentation methods in domain generalization for the first time. First, we formalize the definition of domain generalization and analyze the role of data augmentation in domain generalization. Second, we propose a new taxonomy that categorizes methods into three classes based on the augmentation objectives: domain-level, image-level, and feature-level augmentation. Third, we compare the experimental results of some data augmentation methods on three popular domain generalization datasets and discuss the characteristics and advantages of the current best methods. Fourth, we analyze the shortcomings of each category, propose the suggestions for improvements, and summarize the challenges and the future directions of data augmentation for achieving cross-domain generalization from both theoretical and practical perspectives.

Keywords Data augmentation · Domain generalization · Domain shift · Invariant learning · Model robustness

1 Introduction

Deep learning has gained widespread applications in computer vision, thanks to its powerful learning and representation capabilities. The key to its success lies in two factors: the advancement of powerful GPU devices and the availability of large-scale labeled data [1]. However, obtaining a large amount of labeled data comes with significant costs in terms of manpower and time [2], which is not the ideal pursuit. Moreover, even if we train a high-

✉ Wen'an Zhou
zhouwa@bupt.edu.cn

Yingyi Zhong
zyy2018@bupt.edu.cn

Zhixian Wang
kafkiris@bupt.edu.cn

¹ School of Computer Science, Beijing University of Posts and Telecommunications, Beijing 100876, China

performing deep learning model using richly labeled data, its performance may degrade when there is a distribution shift between the training and testing data [3–5]. In machine learning, this is known as the domain shift problem [6, 7], which can have adverse effects on many real-world applications [8–10].

To address this issue, numerous deep domain adaptation methods have been proposed [11]. Domain adaptation can leverage known source domain data and target data to learn an adaptive model that transfers knowledge from the source domain to the target domain, which can effectively complete the task of assigning labels to unlabeled target domains. However, it would be ineffective when the target data is unseen. In this paper, we discuss the problem of domain generalization [12, 13], which involves both domain shift and the absence of target domain data. Unlike domain adaptation, the goal of domain generalization is to train a model using only one or multiple available source domain datasets, so that the model can be generalized to unseen domains. Existing domain generalization methods can be primarily categorized into three types [12]: domain alignment [14], meta-learning [15], and data augmentation [16].

The data augmentation methods can increase the data available during the training phase and prevent network overfitting by increasing the diversity of the data [17, 18], enabling the model to capture unbiased domain-invariant features and thus facilitating the generalization ability of the model. There have been a lot of surveys about data augmentation. [19] categorized data augmentation into methods based on basic image manipulations and methods based on deep learning, but they blurred the boundary between domain augmentation and image augmentation. [20] pointed out that domain shift was one of the challenges in current computer vision tasks, but they only discussed this issue from the perspective of image style, and the summarized methods were limited to style-altering strategies. [21] classified data augmentation methods into classical image data augmentation and deep learning data augmentation, but their summary was not comprehensive and thorough enough, lack of self-supervised learning and so on. [22] provided a detailed summary of current data augmentation methods, but they did not analyze generalization from a cross-domain perspective. [12] and [13] are surveys of domain generalization, where they briefly mentioned some common data augmentation methods applied in domain generalization.

In this paper, we systematically and comprehensively study data augmentation methods applied in domain generalization. Specifically, Our contributions are as follows:

- We provide a detailed analysis of the role of data augmentation in achieving cross-domain generalization.
- We investigate the development of data augmentation methods in domain generalization and categorize them into three classes based on the augmentation objectives.
- We analyze the current challenges of data augmentation in domain generalization from both theoretical and practical perspectives, and discuss possible future directions.

The remainder of this survey is organized as follows. In Sect. 2, we introduce domain generalization through a formal definition and analyze the role of data augmentation in cross-domain generalization. In Sect. 3, we propose a new taxonomy of data augmentation in domain generalization. In the following three sections, we dive deep into these three methods separately. In Sect. 7, we introduce four widely employed datasets in domain generalization and showcase the comparative results on three of these datasets using the algorithms mentioned in the preceding three sections. The challenges and possible future directions are discussed in Sect. 8. Finally, the conclusion of this survey is given in Sect. 9.

Table 1 The notation and denotation

$\mathcal{X}$	Input space
$\mathcal{Y}$	Output space
P_{XY}	Joint probability distribution
$(\mathbf{x}, \mathbf{y}) \sim P_{XY}$	The samples $(\mathbf{x}, \mathbf{y})$ follow the P_{XY} distribution
$\mathcal{S}, \mathcal{T}$	Source and target domains
n^i	Number of samples in the i -th source domain
M	Number of source domains
n^{M+1}	Number of samples in the target domain

2 Background

2.1 Notations and Definitions

We first introduce some notations and definitions used in this paper. The notations and definitions mainly refer to [12, 13]. In machine learning, a domain is composed of input space $\mathcal{X}$, output space $\mathcal{Y}$, and joint probability distribution P_{XY} . The input space $\mathcal{X}$ and the output space $\mathcal{Y}$ represent respectively the set of all possible inputs to the model and the set of all possible outputs to the model, that is, for $\forall (\mathbf{x}_i, \mathbf{y}_i)$ with $\mathbf{x}_i \in \mathcal{X}$ and $\mathbf{y}_i \in \mathcal{Y}$. For the samples $(\mathbf{x}, \mathbf{y})$ sampled from the distribution P_{XY} , we denote it as $(\mathbf{x}, \mathbf{y}) \sim P_{XY}$, indicating that $(\mathbf{x}, \mathbf{y})$ obey the P_{XY} distribution. We use $\mathcal{D}$ to represent a domain, then $\mathcal{D} = \{\mathcal{X}, \mathcal{Y}, P_{XY}\}$. In domain generalization, we hope to learn generalizable knowledge from one or more source domains available, so that the model can perform well on previously unseen target domains. We define M labeled source domains $\mathcal{S} = \{\mathcal{S}_1, \dots, \mathcal{S}_M\}$, $M \geq 1$ where the i -th source domain $\mathcal{S}_i = \{(\mathbf{x}_k^i, \mathbf{y}_k^i)\}_{k=1}^{n^i} \sim P_{XY}^{(i)}$ contains n^i training samples, $1 \leq i \leq M$ and there are n^{M+1} samples in the target domain $\mathcal{T} = \{(\mathbf{x}_j)\}_{j=1}^{n^{M+1}} \sim P_{XY}^t$ that is unlabeled and unseen during the training phase. The joint probability distributions of the source domains are different: $P_{XY}^{(i)} \neq P_{XY}^{(j)}$, $i \neq j$ and $i, j \in [1, M]$. In general, we assume that the joint probability distribution of the unseen target domain is also different from that of the M source domains: $P_{XY}^t \neq P_{XY}^{(i)}$, $i \in [1, M]$. Table 1 lists all the symbols used in this paper along with their descriptions. The goal of domain generalization is to use the available source domains $\mathcal{S}$ to learn a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ through empirical risk minimization [23], so that it can maintain similar performance on the unseen target domain $\mathcal{T}$. In other words, we aim to obtain a model with strong generalization ability and robustness, which learns transferrable knowledge from the source data and can effectively adapt to new domains with different data distributions from the source domains.

2.2 The Role of Data Augmentation in DG

As mentioned in previous studies [19, 22], data augmentation has been shown to achieve the effect of regularization [24, 25], which helps in mitigating overfitting. Through data transformation (including in the input space [26, 27] and in the embedding space [28, 29]) or data generation (e.g., [30]), the quantity and quality of training data can be effectively increased to help the model learn more complete and stable knowledge [31]. We analyze the

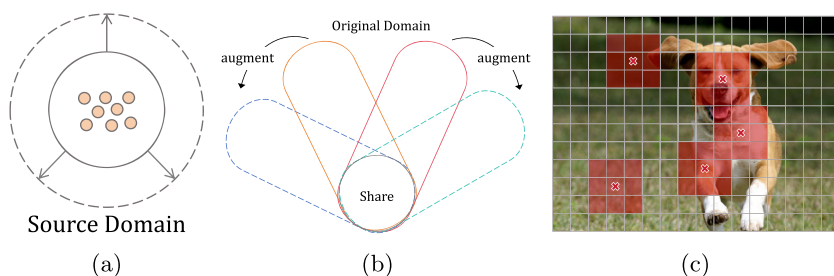


Fig. 1 Illustration of the role of data augmentation

role of data augmentation in detail from the perspective of cross-domain generalization and provide illustrations for each role of data augmentation (as shown in Fig. 1).

Expanding the coverage of the domain Data augmentation generates new data that is similar to the original data but has different distributions, thereby creating a larger and more diverse training dataset. This approach not only increases the likelihood of covering the data distribution of the target domain but also encourages the model to learn more robust features by exposing it to a wider range of possible distributions, thereby improving the robustness of the model.

Strengthening the ability of the model to learn shared features Among the multiple source domains used for training, each domain consists of both domain-private and domain-shared features [32]. Data augmentation, by simulating possible and label-preserving variations in unseen domains, not only introduces diverse data distributions, but also transfers domain-invariant information to the model as prior knowledge. This promotes the model in better distinguishing between domain-private and domain-shared features, which is crucial for learning shared features that are beneficial for the task at hand. Mining shared features from multiple domains is beneficial to improving the robustness of the model.

Avoiding shortcut learning Shortcut learning [33] refers to the model's tendency to learn superficial associations in the training data while overlooking discriminative features relevant to the task. This phenomenon leads to poor generalization performance since the spurious correlations learned by the model may not hold true in new domains. For example, if we train a model to recognize dogs using a large dataset of dogs in the desert, shortcut learning may cause the model to associate desert features (e.g., background) with dog discrimination. As a result, when presented with images of dogs in water during testing, the model may yield unsatisfactory results. Data augmentation can address this issue by explicitly imposing regularization constraints on the model, such as injecting noise [34, 35]. This encourages the network to focus on every detail of the input instead of relying solely on a small subset of visual features, thereby improving the robustness of the model.

3 Taxonomy

Currently, most surveys only focus on discussing data augmentation [19–22] or domain generalization [12, 13], but few of them discuss methods that combine the two. To better understand the usage and impact of data augmentation in domain generalization (DG), we categorize existing methods into three groups based on the augmentation objectives, as shown in Fig. 2.

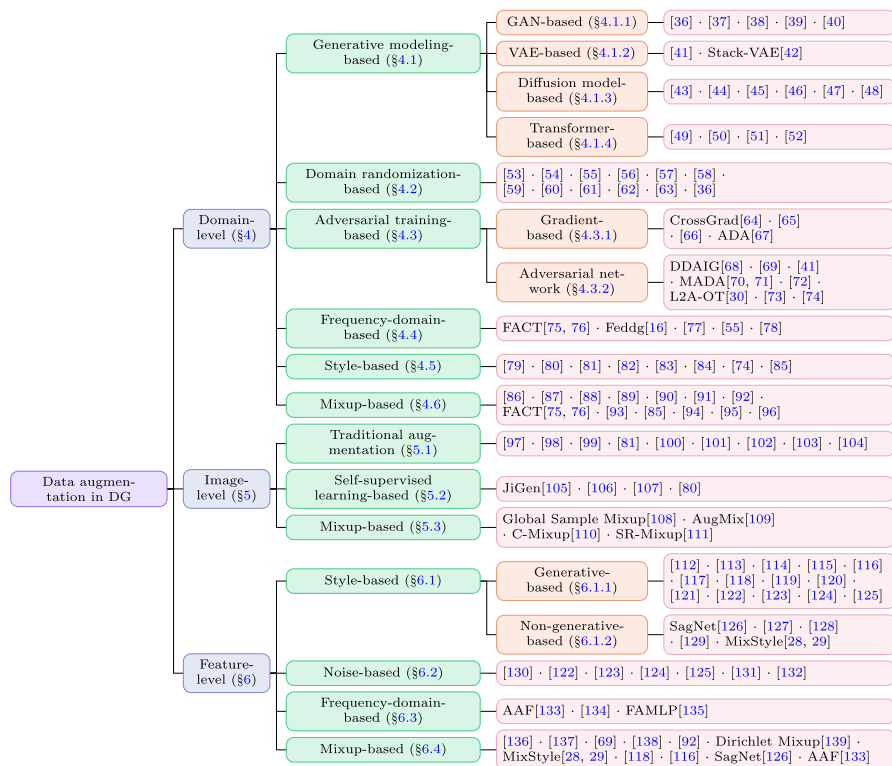


Fig. 2 Taxonomy of data augmentation in DG

Domain-level augmentation is a popular type of data augmentation method applied in domain generalization. Its motivation is simple: to increase the breadth of the training domains and cover the unseen target domain as comprehensively as possible by generating a large number of new samples with diverse distributions. This enables the model to see more diverse data or data similar to the target domain in advance, reducing the impact of domain shift during testing. We provide a detailed introduction to these methods in Sect. 4.

Image-level augmentation creates new examples similar to the original images but does not explicitly focus on the concept of domain. These new examples are created by performing a series of transformations or noise addition to the original images in the training set. While the generation of these examples does not receive additional support from prior knowledge about the domain, it implicitly creates new domains. For example, the definition of domains in [41] refers to the diverse distributions of objects appearance resulting from either external conditions (e.g., weather) or intrinsic attributes (e.g., color). By applying jitter to images, we modify the intrinsic attributes of the target object (such as brightness, contrast, etc.). Consequently, we can effectively enrich the source training domains. These methods will be thoroughly explained in Sect. 5.

Feature-level augmentation refers to augmenting sample features in the embedding space. Unlike the aforementioned two augmentation methods, feature-level augmentation is advantageous as it does not incur significant computational and time costs, nor does it require the design of specialized augmentation schemes for specific tasks or domains, thereby providing greater flexibility. Furthermore, feature-level augmentation has the potential to obtain richer

Table 2 Categorization of domain-level augmentation (In the application scenarios, image classification is included by default)

	Highlight	Application scenarios
Generative modeling-based (Sect. 4.1)	Utilizing existing generative models such as GANs, VAEs, and their variants to generate samples from unseen domains	Autonomous driving [40], semantic segmentation [39]
Domain randomization-based (Sect. 4.2)	Using methods such as computer simulators to introduce randomness and variation into training set	Pose estimation [57], robotics [58], person re-identification [59]
Adversarial training-based (Sect. 4.3)	Employing gradient adversarial training or adversarial networks to generate challenging samples for the model	Traffic scene segmentation [41], text classification [69]
Frequency-domain-based (Sect. 4.4)	Making use of prior knowledge in frequency domain (e.g., phase contains semantic information) to assist augmentation	Federated learning [16], medical image analysis [86]
Style-based (Sect. 4.5)	Using the fact that feature statistics contain style information to change image style	Semantic segmentation [82], remote sensing image processing [141]
Mixup-based (Sect. 4.6)	Performing linear interpolation on samples in different domains	Medical image analysis [142]

and more meaningful semantic information by designing specific feature transformations, as semantic information is more densely packed [20] and certain laws exist in the deep feature space [24, 140]. These methods will be comprehensively explained in Sect. 6.

4 Domain-Level Augmentation

Domain-level augmentation methods can be categorized into six classes: generative modeling-based, domain randomization-based, adversarial training-based, frequency-domain-based, style-based, and Mixup-based. We list these six methods in Table 2, along with a brief summary description.

4.1 Generative Modeling-Based Domain-Level Augmentation

In machine learning, generative models are employed to learn probability distributions from training data using neural networks, and generate new samples from the learned distributions.

In this section, we focus on the deep generative model, which is a type of deep neural network capable of learning complex high-dimensional probability distributions from data. In other words, a deep generative model is designed to learn an unknown or intractable probability distribution that the given few samples obey, and then the model can leverage this knowledge to create new samples similar to those from the new unknown distribution. Deep generative models have been effectively applied for a long time in various tasks, including image compression [143], video generation [144], anomaly detection [145], etc., due to their powerful generation and representation capabilities. Specifically, in domain generalization tasks, deep generative models are employed to synthesize numerous new images from images in a given domain, which resemble the provided images and comply with the data distribution of other specified domains. Obviously, the generated samples of different distributions enrich the initial training datasets, helping to mitigate the influence of domain factors on the model and reduce the model's vulnerability to unknown domains.

In this section, we categorize deep generative modeling methods in domain generalization into four major categories relevant to generative adversarial networks (GAN) [146], variational autoencoders (VAE) [147], diffusion models [148], and transformers [149], which are introduced in detail as follows.

4.1.1 GAN-Based Methods

GAN [146] is an adversarial-based data generation model, consisting of a generator and a discriminator. The goal of the generator is to generate images similar to the input sample through random noise, while the discriminator's goal is to distinguish whether the generated results are from real samples or from the generator's forgery. The generator strives to generate images that are more similar to real samples, while the discriminator strives to recognize the source without being deceived by the generator. The two players alternately optimize each other, and the generator trained in this way can ultimately generate images that are indistinguishable from real ones.

Yue et al. [36] assumed that neural networks could effectively generalize to new, real-world domains if they were exposed to a diverse set of domains during training, so they selected auxiliary domains from the real world and applied CycleGAN [150] to randomize the source domain images into multiple styles of real images. Rahman et al. [37] used ComboGAN [151] to generate samples that were not from the same domain as the original training data, and used these samples as validation data during the training phase. Considering that there were multiple available source domains, [37] also designed a discriminator for each domain-to-domain transformation network, and used adversarial loss and cycle-consistency loss to constrain the differences between the transformation networks. In more detail, Nguyen et al. [38] argued that to achieve domain generalization, it was necessary to maintain marginal and conditional consistency between domains, which meant that both marginal and conditional distributions had to be aligned to learn domain-invariant representations. It was theoretically proven in [38] that both types of consistency could be simultaneously satisfied by using an invertible and differentiable function such as StarGAN [152] as the domain density transformation function.

4.1.2 VAE-Based Methods

VAE [147] is a type of generative model that employs the autoencoder architecture. The main idea is to introduce latent variables that enable the original data to be encoded into a

probability distribution in a latent space. The decoder then maps randomly sampled latent variable vectors back to the sample space to generate new data similar to the original data. Instead of directly mapping the input data to a single point in the latent space, VAE's encoder maps it to a Gaussian distribution in the latent space. In this way, latent space information can be expressed using the parameters of Gaussian distribution, leading to improved sampling and generation of data. Furthermore, to ensure the quality of the generated data, VAE employs a reconstruction loss to encourage the generated data to be as close to the input as possible, and also uses the KL divergence loss as a regularization term to limit the divergence of the latent space.

In order to solve the single-source domain generalization problem, Li et al. [41] proposed a framework called progressive domain expansion network (PDEN), which used a generator such as VAE to continuously synthesize unseen domains to supplement the source domain. Xie et al. [42] highlighted that in zero-shot learning, the model was only supervised by seen classes, making the recognition of unseen classes easily affected by domain shift. To address this challenge, a novel technique called Stack-VAE was proposed in [42], which consisted of a hierarchical VAE generator that utilized multi-layer Gaussian distribution to improve expressiveness, thereby better simulating the real data distribution of the unseen classes.

4.1.3 Diffusion Model-Based Methods

The diffusion model [148] is one of the most popular generative models in recent years, which can generate realistic, high-quality, and diverse augmented images by continuously adding noise to the training image and then learning its inverse denoising process. Compared with other image data augmentation methods, diffusion models are better at learning the underlying data distribution, capturing complex relationships in high-dimensional image data, and preserving the semantic structure of the original data [153]. In addition, utilizing additional knowledge such as class information and text descriptions to guide the data generation process of diffusion models can synthesize images that are more beneficial for downstream tasks [43, 44].

Dunlap et al. [43] proposed an automated language-guided image augmentation method (ALIA), which first used the large language model to automatically create domain descriptions, and then used the diffusion model to perform language-guided image editing. ALIA not only retained the task-relevant information in the original data, but also significantly increased the diversity of the domain, thereby improving the model generalization performance. Noori et al. [44] proposed a Feedback-guided domain synthesis method (FDS), which trained the diffusion model guided by domain and class information in the original dataset, and then combined domain interpolation to generate images with rich diversity and spanning a broad distribution spectrum. Wu et al. [45] proposed a generalized dataset generation model DatasetDM, which used the pre-trained diffusion model to generate an unlimited quantity of high-fidelity synthetic images, and then used the perception decoder to generate corresponding high-quality perception annotations. This approach provided rich and high-quality datasets for a wide range of downstream tasks, significantly improving model generalization and robustness. Feng et al. [46] proposed DiffTPT for test-time prompt tuning, which not only utilized diffusion models to generate augmented data that preserved the key semantics, but also filtered out spurious augmentations through a cosine similarity-based filtration technique, thereby ensuring both data diversity and prediction fidelity. Hemati et al. [47] proposed a cross domain generative augmentation method (CDGA), which utilized the pre-trained latent diffusion model to synthesize images, significantly improving the out-of-distribution generalization ability of vanilla ERM. Niemeijer et al. [48] proposed a diffusion-based domain

extension method (DIDEX) that used the diffusion model to generate images with high diversity. These images were regarded as pseudo target domains, thereby achieving model generalization by solving domain adaptation tasks.

4.1.4 Transformer-Based Methods

Text generation is one of the most important tasks in natural language processing, and transformers [149] have shown great development prospects in this task. Benefiting from the powerful expressing capabilities of the self-attention mechanism, transformer can fully model the dependencies between each input sequence element in the encoders and decoders, generating more fluent and more accurate text. Transformer-based pre-trained language models, such as BERT [154] and GPT [155], deeply learn language knowledge by pre-training on large-scale unsupervised corpora, thereby generating higher quality and more diverse text for downstream tasks.

Amin-Nejad et al. [49] used the transformer-based text generation model to generate structured patient information to augment medical datasets. Bayer et al. [50] designed a GPT-based text data augmentation method to improve the diversity of augmented data while maintaining label quality. Chen et al. [51] proposed a data augmentation method designed for named entity recognition, using GPT for prompt-based data generation to obtain diverse and high-quality training data. Cohen et al. [52] used GPT-3 to rephrase and back-translate sentences during training and testing to assist model training, improving the model's performance across various hate speech detection datasets and metrics.

4.2 Domain Randomization-Based Domain-Level Augmentation

Domain randomization [53, 156] is a widely used data augmentation technique in computer vision that helps reduce the domain gap between synthetic and real data [54]. Methods such as computer simulators [157], image-to-image translation techniques [55], random neural networks [56], and more can be employed to introduce randomness and variations into samples within a simulated environment. This enables the simulation of diverse changes that occur in the real world, ultimately facilitating the generalization of models to real-world environments. For instance, in autonomous driving, factors such as road conditions, weather, and time of day can significantly impact the safety of driving, making it challenging to predict real-world road situations. Domain randomization enables the model to enhance its adaptability to complex scenes by exposing it to diverse variations generated randomly in the simulation environment, thereby improving driving safety.

In domain generalization, data augmentation via domain randomization can generate training data with rich variation to increase the robustness of the model to high environmental variability [60]. Prakash et al. [61] pointed out that when objects in an image only occupied a few pixels, it was important to consider the surrounding background context. Based on this setting, they proposed a structured domain randomization method in [61], which could consider the structure of the scene during data generation and encourage the model to make better use of context when performing tasks. Taking into account that randomly transforming the entire spectrum of images in the spatial space may lead to the loss of domain-invariant features, Huang et al. [55] proposed a technique called frequency space domain randomization (fsdr) to achieve domain generalization by randomly modifying domain-variant frequency components without changing domain-invariant frequency components. To achieve colour-agnostic hand localization in cluttered industrial environments, Grushko et al. [62] trained

the model with a fully synthetic dataset from domain randomization techniques to alleviate the problem that the model was easily degraded due to training on real-world data. In particular, we regard random convolutions [63] as a unique domain randomization technique capable of generating an infinite number of new domain images that maintain the global shapes while introducing random distortions to the local textures. It is worth noting that deep generative models can serve as a solution for image-to-image translation tasks, which implies that the methods mentioned in Sect. 4.1 can also be considered for accomplishing domain randomization (e.g., [36]).

4.3 Adversarial Training-Based Domain-Level Augmentation

Adversarial training-based data augmentation methods aim to modify the original data to generate adversarial samples with different data distributions. These adversarial samples, being challenging for the model to comprehend, are beneficial for identifying deficiencies in the model's learned knowledge. We categorize adversarial training augmentation methods into two types: gradient-based adversarial methods and adversarial networks, which are discussed in detail below.

4.3.1 Gradient-Based Adversarial Methods

Inspired by adversarial attacks [158], most gradient-based adversarial augmentation methods aim to perturb original images by utilizing adversarial gradients from a domain classifier to generate new samples, while ensuring the semantic consistency of the augmented samples by incorporating a task classifier as an adversarial constraint. Following the definition in Sect. 2.1, we assume that the current input to the network is the k -th image in the i -th domain, denoted as (x_k^i, y_k^i, d^i) , and $\mathcal{L}_d(x_k^i, d^i)$ represents the loss of this sample on the domain classifier. Based on $\mathcal{L}_d(x_k^i, d^i)$, we aim to perturb the original image in the direction of gradient ascent, such that the new sample $(x_k^{i'}, y_k^{i'})$ is expected to be as far away as possible from domain d^i , while preserving the label of the class:

$$x_k^{i'} \leftarrow x_k^i + \epsilon \nabla \mathcal{L}_d(x_k^i, d^i), \quad (1)$$

where ϵ is the learning rate.

CrossGrad [64] designed a domain classifier and a label classifier, which alternated data augmentation and parameter optimization during training, utilizing the loss of perturbed samples from the domain classifier to assist in the optimization of the label classifier. Furthermore, ADA [67] considered the setting of single-source domain generalization and added a regularization term to compute the loss of the domain classifier. Inspired by [67, 159], [66] considered augmenting data by solving worst-case problems to facilitate the model in learning more robust inductive biases. In contrast to the aforementioned adversarial methods that maximize classification errors, [65] perturbed the source distribution by maximizing entropy to improve the model's robustness.

4.3.2 Adversarial Network

The purpose of this type of method is to train a neural network that includes a minimax two-player game, inspired by GAN but different. Specifically, GAN serves as a complete data generation framework by utilizing a generator and a discriminator to generate additional

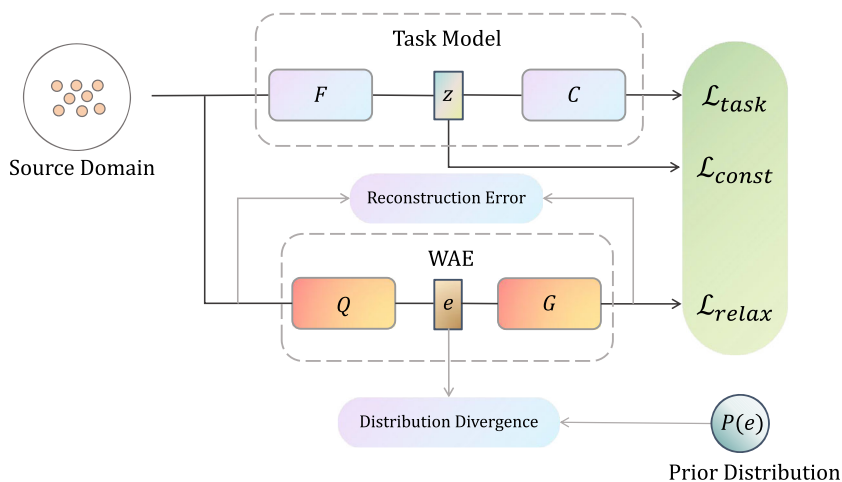


Fig. 3 Overview of adversarial domain augmentation in MADA

training samples. However, in adversarial network-based augmentation methods, we can identify two designs for adversarial learning throughout the network structure: one for data generation and the other for adversarial constraints. Certainly, as mentioned in Sect. 4.1, deep generative models can be utilized as data generators in such methods. However, here we primarily focus on how adversarial network frameworks are designed and how adversarial constraints are incorporated in different studies.

DDAIG [68] trained a domain transformation network to synthesize data from unknown domains, which were then used to assist in learning a domain classifier. Specifically, the domain transformation network aimed to generate samples that preserved reliable semantic information for easy recognition by the label classifier while confusing the domain classifier as adversarial constraints. MADA [70, 71] employed an autoencoder in the input space to generate poor samples to relax the sample space, while also applying an adversarial constraint on the features in the embedding space to ensure semantic consistency (Fig. 3). Similar to MADA, L2A-OT [30] aimed to expand the input space by learning a conditional generator network that maximizes the optimal transport [160] between the source domain and the pseudo-novel domain. It also utilized cycle-consistency loss [150] and classification loss to maintain structural and semantic consistency. Apart from [30], there are other methods similar to MADA in terms of their ideas, such as [72, 73]. Considering contrastive learning, [41] minimized the contrastive loss to ensure that the generated samples with different domains but the same semantics were closer in the latent space, while maximizing this contrastive loss to ensure that the generator generates more effective samples in the unseen domain. Wang et al. [74] introduced a style-complement module to learn the synthesis of samples with unseen styles that lie outside the original distributions. During training, they minimized the tractable upper bound of mutual information between the synthesized and source images, while maximizing the mutual information between images belonging to the same semantic category. By doing so, a min-max game was established between the style-complement module and the task model. In particular, Qiao et al. [69] considered the uncertainty of augmentations and relaxed the constraint through adversarial domain augmentation, allowing the model to learn gradually from "easy" to "hard" domains.

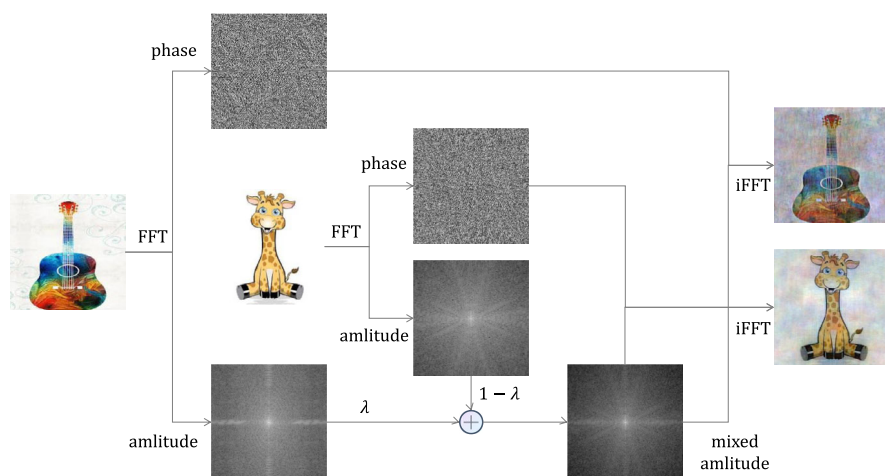


Fig. 4 Illustration of amplitude mix strategy in FACT

4.4 Frequency-Domain-Based Domain-Level Augmentation

The Fourier transform is a technique used to convert an image from the time domain to the frequency domain, enabling the extraction of amplitude and phase information across various frequencies. Many studies [16, 75–77, 161–163] have demonstrated that the phase component of the Fourier spectrum predominantly encapsulates domain-invariant high-level semantics, aiding in mitigating domain shift. On the other hand, the amplitude component encompasses domain-specific low-level statistics that can be altered without affecting the perception of high-level semantics. This property renders the Fourier transform a valuable tool for domain generalization [164].

Inspired by Mixup [165] (for more details, see Sect. 4.6) and the domain adaptation data augmentation method Fda [161], Xu et al. proposed FACT [75, 76] to extract the general domain-invariant features preserved in the phase spectrum by Fourier transforming the input image and linearly interpolating the amplitude spectra of the two images (Fig. 4). In the setting of federated domain generalization, Liu et al. [16] proposed a method where Fourier amplitude information can be exchanged between clients through a continuous interpolation mechanism while preserving phase privacy. Wang et al. [77] disturbed the amplitude and phase by simply applying random noise to them in order to enrich the diversity of the sample distribution. In particular, instead of dealing with amplitude and phase, Huang et al. [55] divided the frequency into domain-invariant frequency components (DIFs) and domain-variant frequency components (DVF) based on their different contributions to feature robustness. They then performed histogram matching [166] on the DVFs of the synthetic image conditioned on the reference real image to obtain the augmented synthetic image.

In addition to the property mentioned above, Some studies [167–169] showed that adversarial noise perturbations were often encoded in high-frequency components, while style information was encoded in low-frequency components. Building upon this observation, Wang et al. [78] proposed a Factorized Frequency Modification approach, which contained two learnable branches of noise perturbation and style transformation to ensure the hardness and diversity of augmented samples respectively.

4.5 Style-Based Domain-Level Augmentation

Image style has been widely employed to characterize the visual domain, and several studies [170, 171] have discovered that feature statistics are associated with image style. Specifically, style information can be found from the statistics (such as means and standard deviations) of each shallow feature layer. Moreover, Ulyanov et al. [172] have found that replacing batch normalization layers with instance normalization layers can effectively eliminate the style discrepancy among different instances without losing semantic information. Based on the aforementioned findings, Huang et al. [171] proposed an adaptive instance normalization method (AdaIN), which replaced the feature statistics of the content images with the feature statistics of the style images to achieve arbitrary style transfer. To be specific, when we are given a content image and a style image, our objective is to replace the style of the former with the style of the latter, which can be achieved using the following formula:

$$\text{AdaIN}(x, x') = \sigma(x') \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \mu(x'), \quad (2)$$

where x and x' denote the feature maps extracted from the content and style image respectively. The channel-wise mean and standard deviation of x' will act as the scaling parameter and shift parameter of the normalized x to realize the style transfer of x .

In domain generalization, several studies [79–83] have employed the AdaIN method, which allows for the generation of data with novel distributions by altering the style of the source images while preserving their semantic content. For example, Somavarapu et al. [79] conducted inter-source stylization by leveraging the implicit variability within the source domains themselves. Asadi et al. [80] achieved arbitrary domain shifting for each jigsaw puzzle by employing a modified version of the Adaptive Instance Normalization method. Zhou et al. [81] added style augmentation as the third view on the basis of FixMatch [173] to solve the problem of semi-supervised domain generalization. Su et al. [83] used the fine-grained semantic information to guide the network in generating augmented images with more diverse styles, so as to simulate style variations in imaging objects and imaging conditions found in real scenes. In particular, Zhao et al. [82] proposed the use of a style hallucination module to generate diverse source samples. They employed farthest point sampling (FPS) [174] to select C styles from all the source styles as basis styles, allowing them to represent the style space to the utmost extent.

In addition to AdaIN, there are other domain-level style augmentation methods that leverage the fact that feature statistics contain style information. For instance, Zhong et al. [84] proposed an adversarial style augmentation approach for domain generalization semantic segmentation, which continuously improved the results on unseen real domains by dynamically generating hard stylized samples in each training iteration. Besides, Wang et al. [74] iteratively optimized based on mutual information to generate new style samples that complemented the style of the original samples while ensuring semantic consistency.

4.6 Mixup-Based Domain-Level Augmentation

Mixup [165] is a popular data augmentation strategy that directly operates on training data to generate additional training examples, without the need for a generative network. According to the definition in Sect. 2.1, a convex combination of given two random data points is used

to synthesize new training data and the corresponding smooth labels:

$$\begin{aligned}\tilde{\mathbf{x}} &= \lambda \mathbf{x}_i + (1 - \lambda) \mathbf{x}_j, \\ \tilde{y} &= \lambda y_i + (1 - \lambda) y_j,\end{aligned}\tag{3}$$

where $\lambda \sim \text{Beta}(\alpha, \alpha)$, and $\alpha \in (0, +\infty)$ is a hyperparameter that is used to control the degree of interpolation between the sampled two pairs. As a simple and effective data augmentation technique that is orthogonal to any network architecture, Mixup has been widely used in domain generalization to improve the diversity of training samples, which can implicitly enhance invariance to domain shifts.

Domain-level Mixup makes full use of inter-domain information and extends the training distribution through linear interpolation of samples in different domains, so as to promote the model to extract robust features that can resist cross-domain data distribution changes. Zhou et al. [86] trained the segmentation model by simply mixing up randomly sampled samples from different domains as data augmentation. In the setting of heterogeneous domain generalization, Wang et al. [87] proposed two sampling strategies (i.e., Mixup from two domains and Mixup from all domains) to mix up samples across domains. In [88], Wang et al. considered the semi-supervised domain generalization setting and generated samples of additional domains between different source domains (labeled and unlabeled domains) by mixing up the raw labeled data and unlabeled data with accurate pseudo-labels. Similarly, Wang et al. [89] simulated common scenarios in reality to train a universal model for the person re-identification task by mixing up labeled synthetic datasets with unlabeled real-world data. Different from the aforementioned two semi-supervised learning methods, for unlabeled data, Yuan et al. [90] innovatively pseudo-labeled them by dynamically tuning the reliable range and then expanded intra- and inter-domain knowledge based on Mixup. In particular, [91, 92] dealt with the zero-shot domain generalization task by incorporating inter-domain and inter-class Mixups.

Unlike the above-mentioned mixup-based methods that treat all regions in the image equally, there are some methods that decouple the discriminative region and the background region in advance, and then implicitly introduce domain-invariant knowledge by mixing up the background or semantic content. For example, FACT [75, 76] generated augmented images by performing mixup between the amplitude spectra of two images while keeping the phase invariant. Xu et al. [93] further designed four different Mixup strategies for Fourier amplitude and phase based on prior knowledge of domain information and class information. Hong et al. [85] decoupled the content and style of images in advance and created more abundant and robust samples by carefully mixing up content and style. Liu et al. [94] proposed a method called decoupled-mixup, which chose style-, context-, or frequency-aware disentanglement functions to decouple each image according to different needs, and then performed Mixup to suppress noise-prone regions while emphasizing discriminative regions.

Although many studies have shown the effectiveness of mixup through experiments [95, 175–178] or theoretical analysis [179, 180], there are still some issues to be aware of. Firstly, the interpolation of Mixup can often result in the generation of noisy data that is difficult to recognize, thereby compromising the discriminative performance of the model [96, 138]. Secondly, the different activity semantic ranges in different domains can lead to the problem of semantic inconsistency between classes [93, 96]. Thirdly, by ignoring the varying contributions of different regions within an image to the task at hand, there is a risk of incorrect matching between categories and features, which will bring adverse effects on the training model [138].

Table 3 Categorization of image-level augmentation (In the application scenarios, image classification is included by default)

	Highlight	Application scenarios
Traditional augmentation (Sect. 5.1)	Increasing the diversity of image samples through image manipulation or image erasing	Face anti-spoofing [99], medical image analysis [100, 101], reinforcement learning [103, 104]
Self-supervised learning-based (Sect. 5.2)	Designing pretext tasks to facilitate model to learn general features helpful for generalization	Medical image analysis [181]
Mixup-based (Sect. 5.3)	Performing linear interpolation on a pair of samples	Audio signal processing [182]

5 Image-Level Augmentation

We categorize image-level augmentation methods into traditional augmentation, self-supervised learning-based, and Mixup-based methods, which are briefly described in Table 3.

5.1 Traditional Image Augmentation

Traditional image augmentation, with its ease of implementation and strong versatility, has been widely applied in computer vision [97]. It can be divided into two methods: image manipulation and image erasing [22]. Image manipulation methods augment images by adjusting the position of the pixels or altering the pixel values of the image, including cropping, flipping, rotation, translation, color jitter, etc. The method of image erasing modifies the image by removing specific parts of an image and replacing them with a predefined value (typically 0, 255, or the mean value of the entire dataset), including cutout [183], random erasing [27], grid mask [26], etc. We illustrate some methods in Fig. 5.

Traditional image augmentation methods can increase the size of the training dataset and enhance the diversity of image samples, which is beneficial for training feature representations that are invariant to data transformations and improving the generalization performance of models [98]. For example, in [99], negative samples were synthesized using augmentation methods such as color jitter and color mask to train the face anti-spoofing model, achieving the goal of generalizing the model to unseen domains without the use of any real-world attack samples. Zhou et al. [81] tackled the semi-supervised domain generalization problem by designing a multi-view consistency learning framework, including weak augmentations such as random crop and flip, as well as strong augmentations like geometric and color transformations. The domain shift problems faced in digital pathology are mainly changes in color, brightness, and contrast. [100, 101] proved that color stain augmentation (including morphological and color augmentation) had a positive impact on boosting the generalization capability of the model. Ilse et al. [102] mitigated the influence of non-causal factors in domain features on identifying labels by applying color perturbations randomly. Additionally, [103, 104] prevented agents from overfitting in the training environment by applying various label-preserving transformations such as rotation, cutout, flip, etc. in reinforcement learning. It is

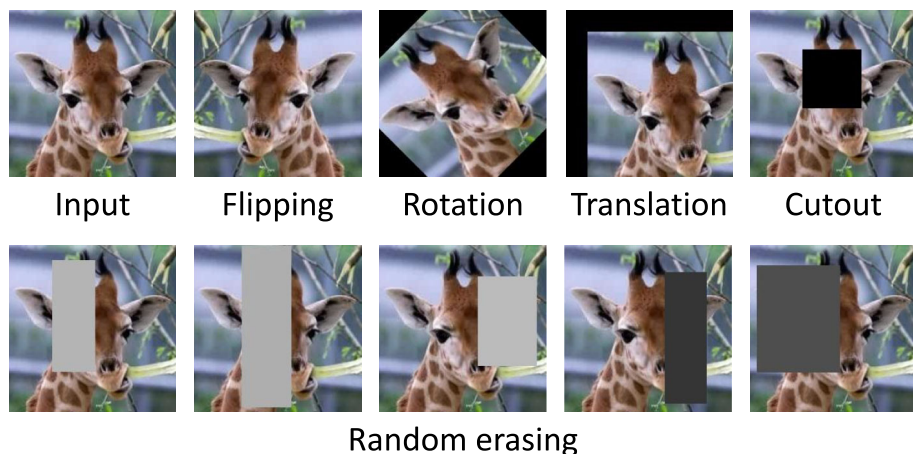


Fig. 5 Some traditional image augmentation methods available in domain generalization

worth noting that inappropriate augmentation schemes may have a negative impact [184], so specific image augmentation schemes should be designed for different tasks.

5.2 Self-supervised Learning-Based Image-Level Augmentation

The major objective of self-supervised learning [185] is to construct supervised information from the original unsupervised data by designing pretext tasks, which aid the model in learning valuable representations that can be utilized for downstream tasks. Zhou et al. [12] pointed out that a model solved pretext tasks without considering the target task, making it easier to learn general features that were helpful for generalization without overfitting to specific domains.

Solving jigsaw puzzles is a common self-supervised task, which is utilized by many researches [80, 105–107] in domain generalization to explore the intrinsic information of objects. We consider solving jigsaw puzzles as an image-level augmentation method for three reasons:

1. It expands the training dataset by randomly shuffling images, thereby increasing the number of training examples available.
2. It enlarges the diversity of samples by altering the spatial structure of the original object, thereby introducing numerous unknown variations into the training data.
3. It enhances the model's comprehension of spatial correlation in objects by challenging the model to correctly order shuffled images, thereby emphasizing the importance of spatial relationships.

Carlucci et al. [105] proposed a jigsaw puzzle-based generalization method called JiGen, in which the network backbone was shared by both the jigsaw task and the classification task. By training the two tasks jointly, the model not only learned the intrinsic regularities of the object that are invariant across domains but also acted as the regularization of the target classification task, enabling the extraction of shared features among multiple domains. Furthermore, Bucci et al. [106] added a pretext task of predicting rotations based on solving jigsaw puzzles. The combination of these two tasks encouraged the model to focus on the global information of images and learn domain-agnostic intrinsic knowledge about objects.

Table 4 Categorization of feature-level augmentation (In the application scenarios, image classification is included by default)

	Highlight	Application scenarios
Style-based (Sect. 6.1)	Perturbing the feature statistics of the original styles to inject different styles in the feature space	Face anti-spoofing [120, 129], semantic segmentation [113], instance retrieval [124]
Noise-based (Sect. 6.2)	Adding noise to the latent features to broaden the feature space of the training domains	Gesture recognition [131], object detection and semantic segmentation [123]
Frequency-domain-based (Sect. 6.3)	Taking full advantage of prior knowledge in frequency domain to assist feature augmentation	Medical image segmentation [133]
Mixup-based (Sect. 4.6)	Performing linear interpolation on the feature vectors	Underwater object detection [136]

Wang et al. [107] utilized metric learning as an external supervision and jigsaw puzzles as an internal supervision to promote the model's understanding of both the relationships among different objects and the objects themselves. Additionally, Asadi et al. [80] alleviated the problem of networks tending to learn superficial statistics of images, such as textures rather than shapes, by combining domain diversification and solving jigsaw puzzles.

5.3 Mixup-Based Image-Level Augmentation

Unlike domain-level mixup, which decouples context and object information or utilizes domain labels to design the mixup strategy, image-level mixup explores new input spaces by mixing up complete images in the original space without considering domain information. When an object experiences significant background changes, the model can effectively capture unbiased semantic features.

Lu et al. [108] proposed the Global Sample Mixup strategy, which leveraged the global information from all samples to improve the mixup variance and promote better decoupling of the background information. Different from the aforementioned method, AugMix [109] injected only one image into multiple augmentation chains and combined the final image generated by the augmentation chains with the original image through random convex combination, thereby ensuring the preservation of both augmentation diversity and semantic information. In particular, Yao et al. [110] proposed C-Mixup as a solution to the regression task, incorporating a mechanism to adjust the sampling probability of mixed up pairs based on the similarity of their labels. Building upon C-mixup, Yasuda et al. [111] further proposed SR-mixup as a solution to the super-resolution problem.

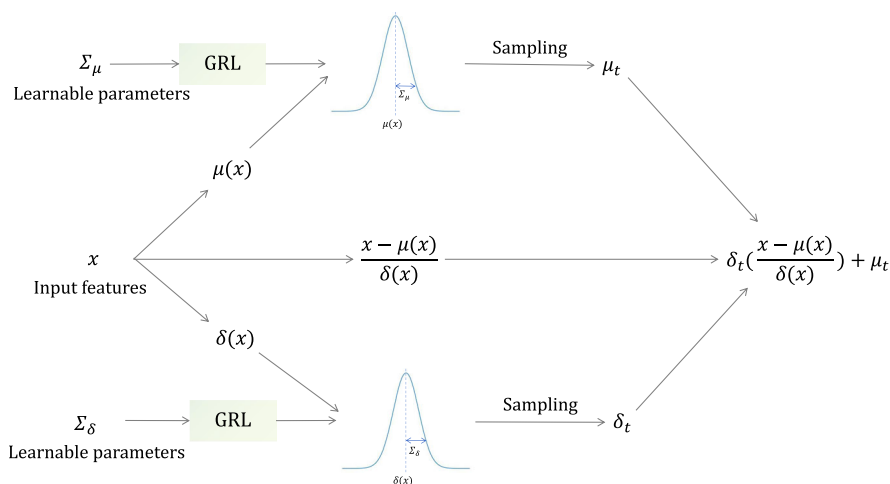


Fig. 6 The adversarial style augmentation module

6 Feature-Level Augmentation

Feature-level augmentation includes style-based, noise-based, frequency-domain-based, and Mixup-based methods, which are briefly presented in Table 4.

6.1 Style-Based Feature-Level Augmentation

According to the characteristics that feature statistics can express style information (i.e., domain information), there are also many methods to implement style augmentation in the feature space by injecting different styles into the feature statistics of the original styles. Depending on whether additional novel styles are generated to supplement the original training domains, we classify feature-level style augmentation methods into generative-based and non-generative-based.

6.1.1 Generative-Based Style Augmentation

This type of approach involves generating a more diverse range of styles to expand the style space. Subsequently, it utilizes the feature statistics of these new styles to perturb the feature statistics of the original styles, thereby achieving style augmentation.

Considering the potential uncertainties in the unknown domain, [112, 113] hypothesized that the feature statistics followed a multivariate Gaussian distribution. They then randomly sampled feature statistics variants from the estimated Gaussian distribution and replaced the original deterministic values to alleviate domain shift. Zhang et al. [114] further considered modeling standard deviations of Gaussian distributions as learnable parameters (Fig. 6) and augmented styles with adversarial training to encourage the model to generalize to the worst-case domain. More finely, Zhang et al. [115] considered the correlation between different feature channels and sampled the correlated feature statistics from the calculated Gaussian distribution. Tan et al. [116] also used Gaussian distributions to model the feature statistics and sampled a new style independent of the original style from them. Furthermore, Chen

et al. [117] added the sensitivity loss to balance the requirements of target feature domain-invariance and sufficient domain feature variations.

In addition to sampling new style feature statistics from the Gaussian distribution, there are other methods to generate additional feature statistics. Kim et al. [118] generated new styles with prediction disturbance, plausibility, and style diversity using Randomized adversarial style perturbations (RASP) and maintained knowledge from source domains using the normalized feature mixup (NFM) technique. Li et al. [119] also increased the local-domain data diversity by perturbing the feature statistics of each domain instance. Zhou et al. [120] selected basis styles of each class by farthest point sampling (FPS) [174], and then linearly combined these basis styles according to the combination weight sampled from Dirichlet distribution to obtain augmented feature statistics. Apart from directly altering the feature statistics of source sample styles, Jeon et al. [121] decomposed the original features into high-frequency (containing structural information) and low-frequency components (containing textural information), and then perturbed the feature statistics of the low-frequency components. There are also several methods that involve adding random noise to generate novel styles [122–125]. We provide a detailed description of these methods in Sect. 6.2.

6.1.2 Non-generative-Based Style Augmentation

This type of approach is committed to utilizing existing styles to introduce the statistics perturbation by performing some techniques such as random swapping or linear interpolation on the feature statistics of different instances.

Nuriel et al. [127] improved AdaIN by exchanging the feature statistics of each instance in a mini-batch with a certain probability during the training phase. Tang et al. [128] used SelfNorm for samples in the same domain to reduce their style discrepancy and employed CrossNorm for samples in different domains to augment styles, which aimed to unify the styles of different domains and improve the robustness of the model. Wang et al. [129] employed a two-stream structure to capture content and style features separately, and they obtained a stylized feature space by sequentially stacking style assembly layers, which assembled randomly shuffled original pairs of content and style features. Differently, Zhou et al. [28, 29] proposed a plug-and-play style augmentation module called MixStyle (Fig. 7), which synthesized images with new styles by randomly mixing up or cross-domain mixing up styles from images of different domains. Similarly, Nam et al. [126] mixed up the feature statistics of the content image with the feature statistics of arbitrary style images to force the content-biased network to pay more attention to domain-invariant content features.

6.2 Noise-Based Feature-Level Augmentation

Many studies have shown that noise injection is an effective data augmentation method in neural networks, which helps to improve the robustness of models [169]. In domain generalization, there are some methods [122–125, 130, 131] to diversify the feature space by adding noise to the latent features to avoid overfitting.

Li et al. [130] broadened the space of the training domains by embedding white Gaussian noise in the latent space to randomly perturb the original feature representations. Wang et al. [122] perturbed the original style statistics by adding random noise to the original feature embeddings. Fan et al. [123] also synthesized diverse potential domains by inserting random noise to perturb the feature channel statistics of original training images, effectively addressing the problem of single-source domain generalization. Kang et al. [124] generated

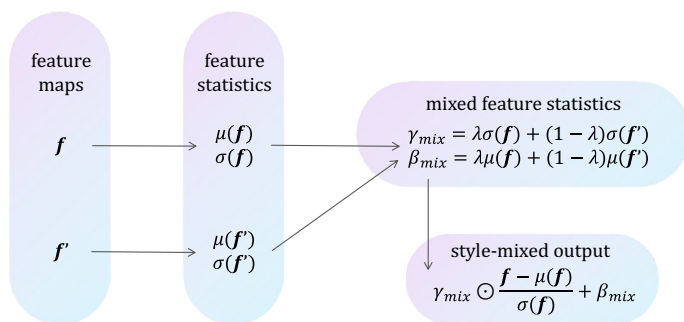


Fig. 7 The calculation process of a mixstyle layer

style candidates by randomly jittering the source style prototypes and then selected novel styles using a given sampling method. Zhang et al. [125] implemented style randomization by perturbing feature statistics with Gaussian noise, and they set different parameters of Gaussian noise in each augmentation process to ensure sufficient style diversity. Wang et al. [131] considered that the random noise might not guarantee the perturbation of features along the class-preserving direction. Therefore, they added an additional label-dependent regularizer to ensure that the key attributes of each class were not modified.

In particular, RSC [132] can also be regarded as a noise-based feature-level augmentation method. RSC iteratively discarded features with a significant contribution to the task and encouraged the network to activate a subset of the remaining features that were also beneficial for prediction. This prevented the model from overfitting to a specific spatial region during training and enabled it to have a higher probability of acquiring a comprehensive understanding of the target object.

6.3 Frequency-Domain-Based Feature-Level Augmentation

Recall from Sect. 4.4 that the Fourier phase captures high-level semantics, while the amplitude represents low-level statistics. In [133], it was pointed out that augmenting the amplitude at feature level had a more pronounced impact compared to augmenting it at image level. This consideration originated from the fact that feature-level augmentation could leverage the high-level style information contained within the multiple channels of features. In the following, we will provide detailed explanations of several approaches that aim to encourage the model to pay more attention to the phase feature by augmenting the Fourier amplitude feature.

In order to better estimate the source style prototype, Zhao et al. [133] proposed a technique called augmenting amplitude features (AAF), which allowed high-level styles to become more diverse by augmenting the amplitude in the early layers of the feature extractor. Hou et al. [134] obtained sample features with different style information from the original data based on the aforementioned Fourier transform characteristics, through (a) performing Fourier transform on the features extracted from different instances, (b) introducing a hyperparameter α to control the proportion of the frequency components that needed to be changed, (c) using Mixup to synthesize new amplitude features, and (d) performing inverse Fourier transform.

Apart from considering the manipulation of phase and amplitude features, the analysis of domain information from the perspective of frequency components has also received attention

[135]. For example, Zheng et al. [135] noted that the domain information was reflected in the high-frequency component of the features, so they designed an adaptive Fourier filter layer to filter the interference of the high-frequency component in the latent features on the domain shift.

6.4 Mixup-Based Feature-Level Augmentation

Feature-level mixup performs linear interpolation on the feature vectors, which, by expanding the searching range in the latent space, facilitates a more comprehensive exploration of diverse representations of the data. Compared to performing Mixup in the input space, feature-level mixup, which leads to a more continuous feature distribution in the latent space with low domain shift, is easier to accommodate the oscillation of the data distribution in the target domain during testing.

Chen et al. [136] divided the features into domain-invariant and domain-specific features, and then performed linear interpolation on the domain-specific features of cross-domain images while keeping the domain-invariant features unchanged. Considering that unrestricted data extrapolation could potentially harm model performance, Peng et al. [137] proposed a method to compute the relation or similarity between instances based on gradients. They then adjusted the weight of mixed up data according to the calculated gradient similarity to ensure meaningful extrapolation. Qiao et al. [69] augmented the input by introducing feature perturbations. Additionally, they proposed a learnable label Mixup technique based on label uncertainty to ensure consistent augmentation in both the input and output spaces.

Moreover, there are also methods that utilize Mixup in the label space to cleverly take advantage of class information to solve target tasks. For common domain generalization task, Lu et al. [138] proposed an extensible DG approach called FIXED, which decreased the intra-class distance by performing mixup on domain-invariant features to facilitate class and domain information disentanglement. Faced with the zero-shot domain generalization task, Mancini et al. [92] generated novel domains and semantic concepts by interpolating among available source domains and categories during training to simulate the domain and semantic shift that could be encountered at test time. To overcome the disparate label sets of open domain generalization, Shu et al. [139] proposed a new Dirichlet mixup to increase domain diversity while compensating for the missing classes in each domain.

Besides, as discussed in Sects. 6.1 and 6.3, methods exist to decompose features into style features and semantic features based on prior knowledge, and then achieve style feature augmentation through mixup. Among these, style-based methods include [28, 29, 116, 118, 126]. And for frequency-domain-based methods, Zhao et al. [133] realized style calibration by mixing up the input amplitude feature with the source prototype.

7 Datasets and Comparison of Methods

Most of the methods mentioned above are based on image classification tasks, so we summarize four datasets used to evaluate image classification tasks in domain generalization in Sect. 7.1. Subsequently, we compare the performance of some of the above methods on different datasets in Sect. 7.2.

7.1 Datasets

In this section, we summarize four popular datasets in domain generalization: PACS [186], VLCS [187], Office-Home [188], and Digits-DG [68].

PACS [186] is a dataset that contains four distinct image styles, with each style corresponding to a domain (Art-painting, Cartoon, Sketch, and Photo). In total, the dataset comprises 7 classes and 9991 images. Domain shift primarily arises from the variations in image styles.

VLCS [187] contains four domains, with each domain originating from four photographic domains: Caltech101, LabelMe, SUN09, and PASCAL VOC 2007. The VLCS dataset comprises 5 classes and 10,729 images totally. Domain shift primarily arises from the variations in image origins.

Office-Home [188] contains four domains (art, clipart, product, and real-world), with a total of 65 classes and 15,500 images. Domain shift primarily arises from the variations in image styles and viewpoints.

Digits-DG [68] consists of four digit datasets: MNIST [189], MNIST-M [190], SVHN [191], SYN [190]. Domain shift primarily arises from the variations in font style, background, and image quality.

7.2 Comparison of Methods

We collate and compare the performance of the aforementioned methods on the PACS, Office-Home, and Digits-DG datasets based on published domain generalization papers, and show the results of each dataset in Table 5 (PACS), Table 6 (office-home), and Table 7 (Digits-DG), respectively. Following previous works [78, 93, 113, 118, 134], all data augmentation methods strictly adhere to the leave-one-domain-out protocol. Specifically, during training, $M - 1$ out of M source domains are used as training data to train the model, while during testing, the remaining domain is used as the target domain to evaluate the model's generalization performance. All experiments use stochastic gradient descent (SGD) as the optimizer. Furthermore, we indicate the backbone network in the table to ensure a fair comparison. More detailed experimental settings follow the papers referenced in the 'Reference' column. Under the same experimental setup, we bold the best result and underline the second best result.

Results on PACS For experiments on PACS, most studies consider using ResNet-18, ResNet-50, and AlexNet as backbone networks. When using ResNet-18 as the backbone network, the best algorithm DSU++ has an average accuracy improvement of 1.1% over the second best algorithm. DSU++ assumed that feature statistics followed a multivariate Gaussian distribution and replaced the original deterministic values with randomly sampled feature statistics from the estimated Gaussian distribution, thereby explicitly modeling the uncertain statistical differences arising from potential domain shifts. When using ResNet-50 as the backbone network, the best algorithm DDC+DIGB improves the average accuracy by 0.6% compared with the second best algorithm. DDC+DIGB first used DDC to mix the amplitude components of two samples, constructing pairs of contrastive samples that shared the same semantic information but differed in domain information. DIGB then guided the model's backpropagation, enhancing the focus on parameters with a consistent optimization direction between pairs of contrastive samples (i.e., domain-invariant parameters). DDC+DIGB effectively captured and updated domain-invariant parameters, improving the extraction of domain-invariant features. When using AlexNet as the backbone network, the best algorithm FFM has an average accuracy improvement of 1.8% over the second best algorithm. FFM

Table 5 Multi-source domain generalization accuracy (%) on PACS

Reference	Methods	Backbone	Photo	Art	Cartoon	Sketch	Avg
[113]	Baseline	ResNet18	96.2	74.3	76.7	68.7	79.0
	Mixup [165]	ResNet18	95.8	76.8	74.9	66.6	78.5
	RSC [132]	ResNet18	94.1	78.9	76.9	76.8	81.7
	L2A-OT [30]	ResNet18	96.2	83.3	78.2	73.6	82.8
	SagNet [126]	ResNet18	95.5	<u>83.6</u>	77.7	76.3	83.3
	pAdaIN [127]	ResNet18	<u>96.3</u>	81.7	76.6	75.1	82.5
	MixStyle [28]	ResNet18	<u>96.3</u>	82.3	79.0	73.8	82.8
	SFA [130]	ResNet18	93.9	81.2	77.8	73.7	81.7
	DSU [112]	ResNet18	95.8	<u>83.6</u>	<u>79.6</u>	<u>77.6</u>	<u>84.1</u>
	DSU++ [113]	ResNet18	96.4	84.5	80.6	79.2	85.2
[134]	Baseline	ResNet50	97.7	86.2	78.7	70.6	83.3
	FACT [75]	ResNet50	96.8	89.6	81.8	<u>84.5</u>	88.2
	RSC [132]	ResNet50	97.9	87.9	82.2	83.4	87.9
	FSDCL [121]	ResNet50	96.6	88.5	83.8	82.9	88.0
	TAF-Cal [133]	ResNet50	<u>97.8</u>	<u>90.5</u>	<u>84.9</u>	83.8	<u>89.3</u>
	DDC+DIGB [134]	ResNet50	97.9	90.8	85.0	85.7	89.9
[78]	JiGen [105]	AlexNet	89.0	67.6	<u>71.7</u>	65.2	73.4
	ADA [67]	AlexNet	85.1	64.3	69.8	60.4	69.9
	MEADA [65]	AlexNet	88.6	67.1	69.9	63.0	72.2
	L2D [74]	AlexNet	91.0	<u>71.2</u>	72.2	<u>67.7</u>	<u>75.5</u>
	FFM [78]	AlexNet	<u>90.8</u>	71.9	71.2	75.3	77.3

highlighted that adversarial perturbation-based data augmentation methods focused on generating hard examples, while style manipulation-based methods aimed to produce diverse samples. Building on this analysis, FFM ensured the diversity and hardness of augmented samples by constructing the noise perturbation branch and the style transformation branch. Additionally, according to the table on PACS, models based on ResNet-50 outperform models based on ResNet-18, while models based on AlexNet show inferior performance.

Results on office-home In experiments on the office-home dataset, most studies consider using ResNet-18 and ResNet-50 as backbone networks. When ResNet-18 is used as the backbone, the average accuracy of the best algorithm DDC+DIGB is 0.5% higher than that of the second best algorithm. This improvement was due to the fact that DDC+DIGB first constructed pairs of contrastive samples with the same semantics through data augmentation, and then mined and updated domain-invariant parameters based on the consistent optimization direction of these contrastive pairs, facilitating the extraction of domain-invariant representations. When using ResNet-50 as the backbone, the average accuracy of the best algorithm RASP+NFM is 2.6% higher than the second best algorithm. RASP+NFM first adversarially augmented feature styles through RASP, and then used NFM for feature normalization to enhance the model's robustness to perturbed features. RASP+NFM not only increased feature diversity through style augmentation but also effectively encouraged the model to maintain source knowledge and improve its adaptation to perturbed features through feature normalization. Additionally, it is evident that models based on ResNet-50 consistently outperform those based on ResNet-18.

Table 6 Multi-source domain generalization accuracy (%) on Office-Home

References	Methods	Backbone	Art	Clipart	Product	Real	Avg
[134]	Baseline	ResNet18	58.9	49.4	74.3	76.2	64.7
	MixStyle [28]	ResNet18	58.7	53.4	74.2	75.9	65.5
	DSU [112]	ResNet18	60.2	54.8	74.1	75.1	66.1
	FACT [75]	ResNet18	60.3	<u>54.9</u>	74.5	76.6	66.6
	FSDCL [121]	ResNet18	60.2	53.5	74.4	76.7	66.2
	TAF-Cal [133]	ResNet18	<u>61.5</u>	55.0	<u>74.9</u>	<u>77.0</u>	<u>67.1</u>
	DDC+DIGB [134]	ResNet18	63.5	52.4	76.0	78.4	67.6
[118]	ERM	ResNet50	64.7	<u>58.8</u>	77.9	79.0	70.1
	CrossGrad [64]	ResNet50	<u>67.7</u>	57.7	<u>79.1</u>	<u>80.4</u>	<u>71.2</u>
	MixStyle [28]	ResNet50	64.9	<u>58.8</u>	78.3	78.7	70.2
	RASP+NFM [118]	ResNet50	68.8	61.7	79.8	80.9	72.8

Table 7 Multi-source domain generalization accuracy (%) on digits-DG (custom backbone is constructed by four 64-kernel 3×3 convolutional layers and a softmax layer. Each convolutional layer is followed by ReLU and 2×2 max-pooling)

References	Methods	Backbone	MNIST	MNIST-M	SVHN	SYN	Avg
[93]	Baseline	Custom	95.8	58.8	61.7	78.6	73.7
	CrossGrad [64]	Custom	96.7	61.1	65.3	80.2	75.8
	DDAIG [68]	Custom	96.6	64.1	68.6	81.0	77.6
	JiGen [105]	Custom	96.5	61.4	63.7	74.0	73.9
	L2A-OT [30]	Custom	96.7	63.9	68.6	83.2	78.1
	FACT [75]	Custom	97.9	<u>65.6</u>	<u>72.4</u>	<u>90.3</u>	<u>81.5</u>
	SAM [93]	Custom	<u>97.8</u>	66.8	73.2	90.7	82.1

Results on digits-DG For digits-DG, most studies consider using a custom CNN backbone network, which is constructed by four 64-kernel 3×3 convolutional layers and a softmax layer. Each convolutional layer is followed by ReLU and 2×2 max-pooling. With the above backbone network, the average accuracy of the best algorithm SAM is improved by 0.6% than that of the second best algorithm. This improvement was due to the fact that SAM considered changes in both semantic space and distribution space when performing data augmentation. By using semantic and domain information to jointly guide the mixup of Fourier phase spectrum and amplitude, SAM effectively reduced the risk of generating large changes and noise.

8 Discussion, Challenges, and Future Directions

Although data augmentation methods in domain generalization have been extensively studied and effectively applied, there are still some challenges that need to be further explored. In this section, we first discuss the shortcomings of domain-level augmentation, image-level augmentation, and feature-level augmentation, and provide the suggestions for improvements (in Sect. 8.1). Then, we analyze the existing challenges and promising research directions

in data augmentation for domain generalization from both theoretical (in Sect. 8.2.1) and practical perspectives (in Sect. 8.2.2).

8.1 Discussion

Domain-level augmentation expands the coverage of the training domain by generating a large number of new samples with varying distributions, thereby reducing the impact of domain shift during testing. This approach emphasizes improving the distribution diversity of augmented samples while maintaining the semantic consistency between augmented samples and original samples. For example, frequency-domain-based and style-based methods use natural semantic rules as prior knowledge to design models that effectively preserve semantic information. Mixup-based methods effectively generate diverse samples. These methods are easy to use but introduce limited diversity. Adversarial training-based methods generate challenging samples but may introduce extreme noise, resulting in unstable training. Compared to other augmentation methods, generative modeling-based methods, especially diffusion model-based and transformer-based methods, generate augmented samples with higher quality and richer diversity, such as more realistic images and more fluent text. However, training these generative models usually consumes a large amount of computational resources. Pre-training generative models on large-scale datasets and then fine-tuning them shows great potential in generating high-quality and diverse augmented samples. In addition, the interpretability of generative models and the controllability of generated samples [192] are also future directions worthy of study.

Image-level augmentation creates augmented samples by applying a series of transformations or adding noise to the original images, enriching the diversity of the training data. This approach is easy to implement, resource-efficient, highly versatile, and can be widely used in different tasks and scenarios. However, image-level augmentation is not guided by domain information, which limits the distribution diversity of augmented samples. In addition, image-level augmentation usually lacks attention to semantic information, which may lead to the introduction of task-irrelevant noise in the augmented samples. These noises may cause the augmented samples to lose the semantics of the original image, thereby increasing the difficulty of model training and reducing model performance. In the future, image-level augmentation should pay more attention to utilizing label information to reduce the generation of task-irrelevant noise. Since inappropriate augmentation schemes may have negative impacts [184], it is worth further exploring how to select specific augmentation schemes for different tasks through automatic data augmentation (introduced in Sect. 8.2.2).

Feature-level augmentation involves transforming embedded representations in the feature space, with a core idea similar to domain-level augmentation—generating more diverse features while preserving semantics. Some feature-level augmentation methods benefit from the development of domain-level augmentation, such as style-based, frequency-domain-based, and mixup-based methods. Compared to augmentation methods in the input space, feature-level augmentation significantly reduces computational and time costs, offers greater flexibility, and is easily compatible with different tasks and model architectures. However, feature-level augmentation can potentially distort features or introduce noise, which may hinder the model's ability to learn cross-domain invariant patterns. Additionally, transformations of features in the embedding space are often difficult to observe and control, which may lead to unpredictable outcomes. Future research will benefit from incorporating stronger semantic consistency constraints while focusing on increasing feature diversity. Adaptive data

augmentation and joint optimization of data augmentation and network training encourage features towards the desired direction of augmentation, as detailed in Sect. 8.2.2.

8.2 Challenges and Future Directions

8.2.1 Theoretical Perspectives

As analyzed in Sect. 2.2, data augmentation, which has the capability to increase the scale and diversity of training data, contributes to improving the model's generalization performance. However, in domain generalization, the use of data augmentation is often based on empirical observations, and the question of **"why data augmentation helps improve the model generalization ability"** remains a thought-provoking issue. Currently, there have been some studies that explain the success of data augmentation methods in cross-domain generalization from a causal perspective [102, 193, 194]. For example, in [102], the researchers simulated causal interventions using data augmentation and demonstrated that it could alleviate the spurious correlation between domains and task labels. [193, 194], on the other hand, explained the effectiveness of counterfactual data augmentation for out-of-domain generalization. Unfortunately, the aforementioned methods could only explain some data augmentation techniques, such as generative modeling and domain randomization, but were unable to explain self-supervised learning and certain noise-based methods (e.g., RSC). In domain generalization, data augmentation methods lack specific theoretical guarantees [13, 22]. In the future, we can consider developing data augmentation theories with a focus on cross-domain generalization from both data and model perspectives. Specifically, **from the perspective of data**, a data augmentation theory should be able to clearly show where the boundary of data distribution should be in order to achieve cross-domain generalization. In other words, the theory should be able to (a) answer how large and diverse the data needed for domain generalization should be, and (b) measure the distance between this distribution boundary and the original data distribution and the augmented data distribution respectively. Furthermore, **from the perspective of the model**, a data augmentation theory should be able to define the relationship between input data and feature importance, i.e., how changes in data affect changes in feature importance. Through this theory, we can calculate the importance of features in the network and measure the changes in importance of features before and after augmentation to provide interpretability for features (using interpretability features can improve the interpretability of the model).

Even if we know why data augmentation is useful, there are still two questions: *"which data augmentation method is most useful in the current task"* and *"how to use a specific data augmentation method most effectively"*. Currently, most data augmentation methods create samples that are not interpretable [195], such as adversarial perturbation and adding noise. It is difficult to judge which method can provide more diversity (we assume that more diverse data can help reduce generalization error), because we lack a method that can measure sample diversity. Just like the analysis we conducted earlier from the perspective of data, a theory should be able to measure the augmented data distribution and answer the question of how far is the current augmented data distribution from the ideal data distribution. It would be helpful in designing appropriate data augmentation schemes for specific problems and providing clear guidance for data augmentation methods to strive towards the most effective directions for improving network performance [196]. Similarly, from the perspective of the model, if a data augmentation theory can characterize the feature importance under different augmentation methods, researchers can choose the most influential augmentation method

based on the strength of feature importance. This can also guide the transformation of data to obtain more discriminative features.

8.2.2 Practical Perspectives

Test-Time Augmentation

Due to the unavailability of the target domain during training, the augmented training data may not fully encompass the diversity and complexity of the target domain. Consequently, all data augmentation methods used during the training phase face this challenge. Based on this, test-time data augmentation methods have gradually gained attention [133]. This approach involves manipulating the testing data to align it more closely with the prediction range where the model performs well, thus reducing the variance in model predictions.

Counterfactual data augmentation

If the spurious correlations present in the data are not taken into account, the model is prone to learning non-causal features that are unrelated to the labels. These non-causal features are unable to address the challenges brought by domain shift. Counterfactual data augmentation [197–199] helps generate counterfactual training samples by incorporating causal relationships as prior knowledge. These augmented samples are then used to train the model, aiming to eliminate the correlation between spurious variables and the prediction variable, thus improving the model's robustness. Furthermore, studying counterfactual data augmentation provides more interpretability and transparency to the model's predictions, facilitating further analysis and adjustments to the model's learning process.

Adaptive data augmentation

Conventional data augmentation methods heavily rely on manual design, and due to the limited and non-directional variations they introduce, they may potentially have negative effects on the task at hand. In contrast, adaptive data augmentation dynamically adjusts the augmentation strategy based on the characteristics of the input data, which is advantageous in guiding the data towards the desired direction of augmentation. For example, in [130], the data adaptive noise generator was devised to ensure that the features are perturbed in the direction of inter-domain variations while preserving the inter-class directions.

Automatic data augmentation

Different from adaptive data augmentation, automatic data augmentation focuses on designing a system or algorithm to automatically search for suitable augmentation strategies for a specific task. For example, AutoAugment [200] utilized reinforcement learning as a searching algorithm and iteratively discovered the optimal policy. Automatic data augmentation offers not only better flexibility and scalability but also produces more efficient samples for target tasks [201], which is worthy of further research and development.

Joint optimization of data augmentation and network training

Data augmentation and network training optimization individually may lead to suboptimal results, as the data augmentation schemes may not explicitly serve the target task. Jointly optimizing data augmentation and network training [202] can yield a task-oriented data augmentation strategy, allowing the target network to learn from it and achieve better performance.

Combining with multi-view learning

Multi-view learning [203–205] aims to improve model performance using data from multiple views, where different views describe the same object from different perspectives. In domain

generalization, semantic-preserving data augmentation generates multiple augmented samples for the original sample, each of which can be regarded as a perspective of the original sample. Combined with multi-view learning, the model can better understand the data and learn domain-invariant features. For example, multi-view clustering [205] encourages the model to learn consistent clustering structures across different views, which is conducive to discovering shared feature patterns across domains, capturing stable structural information, and improving the generalization performance.

9 Conclusion

Data augmentation is a widely used technique in domain generalization, which explicitly adds regularization constraints to improve the model's ability to resist overfitting and generalization. In this study, we formalize the definition of domain generalization and discuss the role of data augmentation from the perspective of cross-domain generalization. Subsequently, we conduct an in-depth analysis of data augmentation methods in domain generalization at three levels: domain-level, image-level, and feature-level. Additionally, we compare the performance of existing data augmentation methods on three popular domain generalization datasets: PACS, Office-Home, and Digits-DG, and discuss the characteristics and advantages of the current state-of-the-art methods. Finally, we analyze the shortcomings of each category, propose the suggestions for improvements, and highlight the challenges and promising future directions of existing methods from both theoretical and practical perspectives. We hope this survey will provide valuable insights to researchers and serve as a source of inspiration, fostering further advancements in the field.

Acknowledgements This work was supported in part by the National Natural Science Foundation of China under Grant No. 62171051.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Sun C, Shrivastava A, Singh S, Gupta A (2017) Revisiting unreasonable effectiveness of data in deep learning era. In: Proceedings of the IEEE international conference on computer vision, pp 843–852. IEEE Computer Society, Venice, Italy
2. Yang X, Che H, Leung M-F, Liu C (2023) Adaptive graph nonnegative matrix factorization with the self-paced regularization. *Appl Intell* 53(12):15818–15835
3. Ben-David S, Blitzer J, Crammer K, Kulesza A, Pereira F, Vaughan JW (2010) A theory of learning from different domains. *Mach Learn* 79:151–175. <https://doi.org/10.1007/s10994-009-5152-4>

4. Recht B, Roelofs R, Schmidt L, Shankar V (2019) Do imagenet classifiers generalize to imagenet?. In: International conference on machine learning, pp 5389–5400. PMLR, Long Beach, California, USA
5. Taori R, Dave A, Shankar V, Carlini N, Recht B, Schmidt L (2020) Measuring robustness to natural distribution shifts in image classification. *Adv Neural Inf Process Syst* 33:18583–18599
6. Pan SJ, Yang Q (2010) A survey on transfer learning. *IEEE Trans Knowl Data Eng* 22(10):1345–1359. <https://doi.org/10.1109/TKDE.2009.191>
7. Torralba A, Efros AA (2011) Unbiased look at dataset bias. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 1521–1528. IEEE Computer Society, Colorado Springs, CO, USA
8. Fan Z, Xu Q, Jiang C, Ding SX (2023) Deep mixed domain generalization network for intelligent fault diagnosis under unseen conditions. *IEEE Trans Industr Electron* 71(1):965–974. <https://doi.org/10.1109/TIE.2023.3243293>
9. Liu H, Song P, Ding R (2020) Towards domain generalization in underwater object detection. In: 2020 IEEE international conference on image processing (ICIP), pp 1971–1975. IEEE, Abu Dhabi, United Arab Emirates
10. Wang R, Chen F, Tang J, Yan P (2022) Adaptive camera margin for mask-guided domain adaptive person re-identification. In: Proceedings of the 30th ACM international conference on multimedia, pp 668–677. ACM, Lisboa, Portugal
11. Wang M, Deng W (2018) Deep visual domain adaptation: a survey. *Neurocomputing* 312:135–153. <https://doi.org/10.1016/j.neucom.2018.05.083>
12. Zhou K, Liu Z, Qiao Y, Xiang T, Loy CC (2023) Domain generalization: a survey. *IEEE Trans Pattern Anal Mach Intell* 45(4):4396–4415. <https://doi.org/10.1109/TPAMI.2022.3195549>
13. Wang J, Lan C, Liu C, Ouyang Y, Qin T, Lu W, Chen Y, Zeng W, Yu P (2023) Generalizing to unseen domains: a survey on domain generalization. *IEEE Trans Knowl Data Eng* 35(8):8052–8072. <https://doi.org/10.1109/TKDE.2022.3178128>
14. Matsuura T, Harada T (2020) Domain generalization using a mixture of multiple latent domains. In: Proceedings of the AAAI conference on artificial intelligence, vol 34, pp 11749–11756. AAAI Press, New York, NY, USA
15. Li D, Yang Y, Song Y-Z, Hospedales T (2018) Learning to generalize: meta-learning for domain generalization. In: Proceedings of the AAAI conference on artificial intelligence, vol 32. AAAI Press, New Orleans, Louisiana, USA
16. Liu Q, Chen C, Qin J, Dou Q, Heng P-A (2021) Feddg: federated domain generalization on medical image segmentation via episodic learning in continuous frequency space. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 1013–1023. Computer Vision Foundation/IEEE, virtual
17. Xie Q, Dai Z, Hovy E, Luong T, Le Q (2020) Unsupervised data augmentation for consistency training. *Adv Neural Inf Process Syst* 33:6256–6268
18. Zhang C, Bengio S, Hardt M, Recht B, Vinyals O (2017) Understanding deep learning requires rethinking generalization. In: 5th international conference on learning representations. OpenReview.net, Toulon, France
19. Shorten C, Khoshgoftaar TM (2019) A survey on image data augmentation for deep learning. *J Big Data* 6(1):1–48. <https://doi.org/10.1186/s40537-019-0197-0>
20. Xu M, Yoon S, Fuentes A, Park DSA (2023) comprehensive survey of image augmentation techniques for deep learning. *Pattern Recognit* 137:109347. <https://doi.org/10.1016/j.patcog.2023.109347>
21. Khalifa NE, Loey M, Mirjalili S (2022) A comprehensive survey of recent trends in deep learning for digital images augmentation. *Artif Intell Rev* 55(3):2351–2377. <https://doi.org/10.1007/s10462-021-10066-4>
22. Kumar T, Turab M, Raj K, Mileo A, Brennan R, Bendechache M (2023) Advanced data augmentation approaches: a comprehensive survey and future directions. *CoRR* <https://doi.org/10.48550/arXiv.2301.02830>
23. Gulrajani I, Lopez-Paz D (2021) In search of lost domain generalization. In: 9th international conference on learning representations. OpenReview.net, Virtual Event, Austria
24. Wang Y, Huang G, Song S, Pan X, Xia Y, Wu C (2022) Regularizing deep networks with semantic data augmentation. *IEEE Trans Pattern Anal Mach Intell* 44(7):3733–3748. <https://doi.org/10.1109/TPAMI.2021.3052951>
25. Kukačka J, Golkov V, Cremers D (2017) Regularization for deep learning: a taxonomy. *CoRR arXiv:1710.10686*
26. Chen P, Liu S, Zhao H, Jia J (2020) Gridmask data augmentation. *CoRR arXiv:2001.04086*

27. Zhong Z, Zheng L, Kang G, Li S, Yang Y (2020) Random erasing data augmentation. In: Proceedings of the AAAI conference on artificial intelligence, vol 34, pp 13001–13008. AAAI Press, New York, NY, USA
28. Zhou K, Yang Y, Qiao Y, Xiang T (2021) Domain generalization with mixstyle. In: 9th international conference on learning representations. OpenReview.net, Virtual Event, Austria
29. Zhou K, Yang Y, Qiao Y, Xiang T (2021) Mixstyle neural networks for domain generalization and adaptation. CoRR [arXiv:2107.02053](https://arxiv.org/abs/2107.02053)
30. Zhou, K., Yang, Y., Hospedales, T., Xiang, T.: Learning to generate novel domains for domain generalization. In: Proceedings of the European Conference on Computer Vision (ECCV), vol. 12361, pp. 561–578. Springer, Glasgow, UK (2020)
31. Hendrycks D, Basart S, Mu N, Kadavath S, Wang F, Dorundo E, Desai R, Zhu T, Parajuli S, Guo M et al (2021) The many faces of robustness: A critical analysis of out-of-distribution generalization. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 8340–8349. IEEE, Montreal, QC, Canada
32. Lu Y, Wu Y, Liu B, Zhang T, Li B, Chu Q, Yu N (2020) Cross-modality person re-identification with shared-specific feature transfer. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 13379–13389. Computer Vision Foundation/IEEE, Seattle, WA, USA
33. Geirhos R, Jacobsen J-H, Michaelis C, Zemel R, Brendel W, Bethge M, Wichmann FA (2020) Shortcut learning in deep neural networks. Nat Mach Intell 2(11):665–673. <https://doi.org/10.1038/s42256-020-00257-z>
34. Ghiasi G, Lin T-Y, Le QV (2018) Dropblock: a regularization method for convolutional networks. Advances in neural information processing systems 31
35. Faramarzi M, Amini M, Badrinaaraayanan A, Verma V, Chandar S (2022) Patchup: a feature-space block-level regularization technique for convolutional neural networks. In: Proceedings of the AAAI conference on artificial intelligence, vol 36, pp 589–597. AAAI Press, Virtual Event
36. Yue X, Zhang Y, Zhao S, Sangiovanni-Vincentelli A, Keutzer K, Gong B (2019) Domain randomization and pyramid consistency: simulation-to-real generalization without accessing target domain data. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 2100–2110. IEEE, Seoul, Korea (South)
37. Rahman MM, Fookes C, Baktashmotlagh M, Sridharan S (2019) Multi-component image translation for deep domain generalization. In: 2019 IEEE winter conference on applications of computer vision (WACV), pp 579–588. IEEE, Waikoloa Village, HI, USA
38. Nguyen AT, Tran T, Gal Y, Baydin AG (2021) Domain invariant representation learning with domain density transformations. Adv Neural Inf Process Syst 34:5264–5275
39. Gong R, Li W, Chen Y, Gool LV (2019) Dlow: domain flow for adaptation and generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 2477–2486. Computer Vision Foundation/IEEE, Long Beach, CA, USA
40. Li Y, Zhang D, Keuper M, Khoreva A (2023) Intra-source style augmentation for improved domain generalization. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 509–519. IEEE, Waikoloa, HI, USA
41. Li L, Gao K, Cao J, Huang Z, Weng Y, Mi X, Yu Z, Li X, Xia B (2021) Progressive domain expansion network for single domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 224–233. Computer Vision Foundation/IEEE, virtual
42. Xie J, Wu J, Liang T, Meng M (2021) Stack-vae network for zero-shot learning. In: Neural information processing: 28th international conference, vol 13111. ICONIP 2021, Sanur, Bali, Indonesia, December 8–12, 2021, Proceedings, Part IV. Springer, Sanur, Bali, Indonesia, pp 250–261
43. Dunlap L, Umuno A, Zhang H, Yang J, Gonzalez JE, Darrell T (2023) Diversify your vision datasets with automatic diffusion-based augmentation. Adv Neural Inf Process Syst 36:79024–79034
44. Noori M, Cheraghali Khani M, Bahri A, Hakim GAV, Osowiechi D, Yazdanpanah M, Ayed IB, Desrosiers C (2024) Feedback-guided domain synthesis with multi-source conditional diffusion models for domain generalization. arXiv preprint [arXiv:2407.03588](https://arxiv.org/abs/2407.03588)
45. Wu W, Zhao Y, Chen H, Gu Y, Zhao R, He Y, Zhou H, Shou MZ, Shen C (2023) Datasetdm: synthesizing data with perception annotations using diffusion models. Adv Neural Inf Process Syst 36:54683–54695
46. Feng C-M, Yu K, Liu Y, Khan S, Zuo W (2023) Diverse data augmentation with diffusions for effective test-time prompt tuning. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 2704–2714
47. Hemati S, Beitollahi M, Estiri AH, Al Omari B, Chen X, Zhang G (2024) Leveraging generative foundation models for domain generalization. In: ICML 2024 workshop on foundation models in the wild

48. Niemeijer J, Schwonberg M, Termöhlen J-A, Schmidt NM, Fingscheidt T (2024) Generalization by adaptation: diffusion-based domain extension for domain-generalized semantic segmentation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 2830–2840
49. Amin-Nejad A, Ive J, Velupillai S (2020) Exploring transformer text generation for medical dataset augmentation. In: Proceedings of the twelfth language resources and evaluation conference, pp 4699–4708
50. Bayer M, Kaufhold M-A, Buchhold B, Keller M, Dallmeyer J, Reuter C (2023) Data augmentation in natural language processing: a novel text generation approach for long and short text classifiers. *Int J Mach Learn Cybern* 14(1):135–150
51. Chen H, Zhang W, Cheng L, Ye H (2022) Diverse and high-quality data augmentation using GPT for named entity recognition. In: International conference on neural information processing, pp 272–283. Springer
52. Cohen S, Presil D, Katz O, Arbili O, Messina S, Rokach L (2023) Enhancing social network hate detection using back translation and GPT-3 augmentations during training and test-time. *information Fusion* 99, 101887
53. Tobin J, Fong R, Ray A, Schneider J, Zaremba W, Abbeel P (2017) Domain randomization for transferring deep neural networks from simulation to the real world. In: 2017 IEEE/RSJ international conference on intelligent robots and systems (IROS), pp 23–30. IEEE, Vancouver, BC, Canada
54. Tremblay J, Prakash A, Acuna D, Brophy M, Jampani V, Anil C, To T, Cameracci E, Bochoon S, Birchfield S (2018) Training deep networks with synthetic data: Bridging the reality gap by domain randomization. In: Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp 969–977. Computer Vision Foundation/IEEE Computer Society, Salt Lake City, UT, USA
55. Huang J, Guan D, Xiao A, Lu S (2021) Fsd: Frequency space domain randomization for domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 6891–6902. Computer Vision Foundation/IEEE, virtual
56. Lee K, Lee K, Shin J, Lee H (2020) Network randomization: a simple technique for generalization in deep reinforcement learning. In: 8th international conference on learning representations. OpenReview.net, Addis Ababa, Ethiopia
57. Sundermeyer M, Marton Z-C, Durner M, Brucker M, Triebel R (2018) Implicit 3D orientation learning for 6d object detection from RGB images. In: Proceedings of the European conference on computer vision (ECCV), vol 11210, pp 712–729. Springer, Munich, Germany
58. Peng XB, Andrychowicz M, Zaremba W, Abbeel P (2018) Sim-to-real transfer of robotic control with dynamics randomization. In: 2018 IEEE international conference on robotics and automation (ICRA), pp 1–8. IEEE, Brisbane, Australia
59. Wang Y, Liao S, Shao L (2020) Surpassing real-world source training data: random 3D characters for generalizable person re-identification. In: Proceedings of the 28th ACM international conference on multimedia, pp 3422–3430. ACM, Virtual Event/Seattle, WA, USA
60. Jackson PT, Abarghouei AA, Bonner S, Breckon TP, Obara B (2019) Style augmentation: data augmentation via style randomization. In: Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp 83–92. Computer Vision Foundation/IEEE, Long Beach, CA, USA
61. Prakash A, Bochoon S, Brophy M, Acuna D, Cameracci E, State G, Shapira O, Birchfield S (2019) Structured domain randomization: Bridging the reality gap by context-aware synthetic data. In: 2019 International conference on robotics and automation (ICRA), pp 7249–7255. IEEE, Montreal, QC, Canada
62. Grushko S, Vysocký A, Chlebek J, Prokop, P (2023) Hadr: Applying domain randomization for generating synthetic multimodal dataset for hand instance segmentation in cluttered industrial environments. *CoRR* <https://doi.org/10.48550/arXiv.2304.05826>
63. Xu Z, Liu D, Yang J, Raffel C, Niethammer M (2021) Robust and generalizable visual representation learning via random convolutions. In: 9th international conference on learning representations. OpenReview.net, Virtual Event, Austria
64. Shankar S, Piratla V, Chakrabarti S, Chaudhuri S, Jyothi P, Sarawagi S (2018) Generalizing across domains via cross-gradient training. In: 6th international conference on learning representations. OpenReview.net, Vancouver, BC, Canada
65. Zhao L, Liu T, Peng X, Metaxas D (2020) Maximum-entropy adversarial data augmentation for improved generalization and robustness. *Adv Neural Inf Process Syst* 33:14435–14447
66. Wang H, Mai H, Gong Y, Deng Z-H (2023) Towards well-generalizing meta-learning via adversarial task augmentation. *Artif Intell* 317:103875. <https://doi.org/10.1016/j.artint.2023.103875>
67. Volpi R, Namkoong H, Sener O, Duchi JC, Murino V, Savarese S (2018) Generalizing to unseen domains via adversarial data augmentation. *Adv Neural Inf Process Syst* 31:5339–5349

68. Zhou K, Yang Y, Hospedales T, Xiang T (2020) Deep domain-adversarial image generation for domain generalisation. In: Proceedings of the AAAI conference on artificial intelligence, vol 34, pp 13025–13032. AAAI Press, New York, NY, USA
69. Qiao F, Peng X (2021) Uncertainty-guided model generalization to unseen domains. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 6790–6800. Computer Vision Foundation/IEEE, virtual
70. Qiao F, Zhao L, Peng X (2020) Learning to learn single domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 12553–12562. Computer Vision Foundation/IEEE, Seattle, WA, USA
71. Peng X, Qiao F, Zhao L (2022) Out-of-domain generalization from a single source: an uncertainty quantification approach. *IEEE Trans Pattern Anal Mach Intell*
72. Yang F-E, Cheng Y-C, Shiau Z-Y, Wang Y-CF (2021) Adversarial teacher-student representation learning for domain generalization. *Adv Neural Inf Process Syst* 34:19448–19460
73. Gokhale T, Anirudh R, Thiagarajan JJ, Kaikhura B, Baral C, Yang Y (2023) Improving diversity with adversarially learned transformations for domain generalization. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 434–443. IEEE, Waikoloa, HI, USA
74. Wang Z, Luo Y, Qiu R, Huang Z, Baktashmotlagh M (2021) Learning to diversify for single domain generalization. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 814–823. IEEE, Montreal, QC, Canada
75. Xu Q, Zhang R, Zhang Y, Wang Y, Tian Q (2021) A fourier-based framework for domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14383–14392. Computer Vision Foundation/IEEE, virtual
76. Xu Q, Zhang R, Fan Z, Wang Y, Wu Y-Y, Zhang Y (2023) Fourier-based augmentation with applications to domain generalization. *Pattern Recognit* 139:109474. <https://doi.org/10.1016/j.patcog.2023.109474>
77. Wang J, Du R, Chang D, Liang K, Ma Z (2022) Domain generalization via frequency-domain-based feature disentanglement and interaction. In: Proceedings of the 30th ACM international conference on multimedia, pp 4821–4829. ACM, Lisboa, Portugal
78. Wang Z, Luo Y, Huang Z, Baktashmotlagh M (2023) Ffm: Injecting out-of-domain knowledge via factorized frequency modification. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 4124–4133. IEEE, Waikoloa, HI, USA
79. Somavarapu N, Ma C-Y, Kira Z (2020) Frustratingly simple domain generalization via image stylization. *CoRR arXiv:2006.11207*
80. Asadi N, Sarfi AM, Hosseinzadeh M, Karimpour Z, Eftekhari M (2019) Towards shape biased unsupervised representation learning for domain generalization. *CoRR arXiv:1909.08245*
81. Zhou K, Loy CC, Liu Z (2023) Semi-supervised domain generalization with stochastic stylematch. *Int J Comput Vis* 131(9):2377–2387. <https://doi.org/10.1007/s11263-023-01821-x>
82. Zhao Y, Zhong Z, Zhao N, Sebe N, Lee GH (2022) Style-hallucinated dual consistency learning for domain generalized semantic segmentation. In: Proceedings of the European conference on computer vision (ECCV), vol 13688, pp 535–552. Springer, Tel Aviv, Israel
83. Su S, Wang H, Yang M (2022) Consistency learning based on class-aware style variation for domain generalizable semantic segmentation. In: Proceedings of the 30th ACM international conference on multimedia, pp 6029–6038. ACM, Lisboa, Portugal
84. Zhong Z, Zhao Y, Lee GH, Sebe N (2022) Adversarial style augmentation for domain generalized urban-scene segmentation. In: Advances in neural information processing systems. http://papers.nips.cc/paper_files/paper/2022/hash/023d94f44110b9a3c62329beec739772-Abstract-Conference.html
85. Hong M, Choi J, Kim G (2021) Stylemix: separating content and style for enhanced data augmentation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14862–14870. Computer Vision Foundation/IEEE, virtual
86. Zhou Z, Qi L, Shi Y (2022) Generalizable medical image segmentation via random amplitude mixup and domain-specific image restoration. In: Proceedings of the European conference on computer vision (ECCV), vol 13681, pp 420–436. Springer, Tel Aviv, Israel
87. Wang Y, Li H, Kot AC (2020) Heterogeneous domain generalization via domain mixup. In: ICASSP 2020–2020 IEEE international conference on acoustics. Speech and signal processing (ICASSP). IEEE, Barcelona, Spain, pp 3622–3626
88. Wang R, Qi L, Shi Y, Gao Y (2023) Better pseudo-label: joint domain-aware label and dual-classifier for semi-supervised domain generalization. *Pattern Recogn* 133:108987. <https://doi.org/10.1016/j.patcog.2022.108987>
89. Wang W, Liao S, Zhao F, Kang C, Shao L (2021) Domainmix: learning generalizable person re-identification without human annotations. In: 32nd British machine vision conference, p 355. BMVA Press, Online

90. Yuan J, Ma X, Chen D, Kuang K, Wu F, Lin L (2022) Label-efficient domain generalization via collaborative exploration and generalization. In: Proceedings of the 30th ACM international conference on multimedia, pp 2361–2370. ACM, Lisboa, Portugal
91. Paul S, Dutta T, Biswas S (2021) Universal cross-domain retrieval: Generalizing across classes and domains. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 12036–12044. IEEE, Montreal, QC, Canada
92. Mancini M, Akata Z, Ricci E, Caputo B (2020) Towards recognizing unseen categories in unseen domains. In: Proceedings of the European conference on computer vision (ECCV), vol 12368, pp 466–483. Springer, Glasgow, UK
93. Xu C, Tian X (2023) Semantic-aware mixup for domain generalization. In: International joint conference on neural networks, pp 1–8. IEEE, Gold Coast, Australia
94. Liu H, Zhang W, Xie J, Wu H, Li B, Zhang Z, Li Y, Huang Y, Ghanem B, Zheng Y (2022) Decoupled mixup for out-of-distribution visual recognition. In: Proceedings of the European conference on computer vision workshops, vol 13806, pp 451–464. Springer, Tel Aviv, Israel
95. Dubey A, Ramanathan V, Pentland A, Mahajan D (2021) Adaptive methods for real-world domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14340–14349. Computer Vision Foundation/IEEE, virtual
96. Lu W, Wang J, Chen Y, Pan SJ, Hu C, Qin X (2022) Semantic-discriminative mixup for generalizable sensor-based cross-domain activity recognition. In: Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies 6(2):1–19. <https://doi.org/10.1145/3534589>
97. Hu J, Gu X, Wang Z, Gu X (2023) Mixture of calibrated networks for domain generalization in brain tumor segmentation. *Knowl-Based Syst* 270:110520. <https://doi.org/10.1016/j.knosys.2023.110520>
98. Chen T, Kornblith S, Norouzi M, Hinton G (2020) A simple framework for contrastive learning of visual representations. In: International conference on machine learning, vol 119, pp 1597–1607. PMLR, Virtual Event
99. Wang W, Liu P, Zheng H, Ying R, Wen F (2023) Domain generalization for face anti-spoofing via negative data augmentation. *IEEE Trans In Forens Secur* 18:2333–2344. <https://doi.org/10.1109/TIFS.2023.3266138>
100. Arvidsson I, Overgaard NC, Marginean F-E, Krzyzanowska A, Bjartell A, Åström K, Heyden A (2018) Generalization of prostate cancer classification for multiple sites using deep learning. In: 2018 IEEE 15th international symposium on biomedical imaging (ISBI 2018), pp 191–194. IEEE, Washington, DC, USA
101. Tellez D, Litjens G, Bándi P, Bulten W, Bokhorst J-M, Ciompi F, Van Der Laak J (2019) Quantifying the effects of data augmentation and stain color normalization in convolutional neural networks for computational pathology. *Med Image Anal* 58:101544. <https://doi.org/10.1016/j.media.2019.101544>
102. Ilse M, Tomczak JM, Forré P (2021) Selecting data augmentation for simulating interventions. In: International conference on machine learning, vol 139, pp 4555–4562. PMLR, virtual event
103. Laskin M, Lee K, Stooke A, Pinto L, Abbeel P, Srinivas A (2020) Reinforcement learning with augmented data. *Adv Neural Inf Process Syst* 33:19884–19895
104. Yarats D, Kostrikov I, Fergus R (2021) Image augmentation is all you need: regularizing deep reinforcement learning from pixels. In: 9th international conference on learning representations. OpenReview.net, virtual
105. Carlucci FM, D’Innocente A, Bucci S, Caputo B, Tommasi T (2019) Domain generalization by solving jigsaw puzzles. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 2229–2238. Computer Vision Foundation / IEEE, Long Beach, CA, USA
106. Bucci S, D’Innocente A, Liao Y, Carlucci FM, Caputo B, Tommasi T (2022) Self-supervised learning across domains. *IEEE Trans Pattern Anal Mach Intell* 44(9):5516–5528. <https://doi.org/10.1109/TPAMI.2021.3070791>
107. Wang S, Yu L, Li C, Fu C-W, Heng P-A (2020) Learning from extrinsic and intrinsic supervisions for domain generalization. In: Proceedings of the European conference on computer vision (ECCV), vol 12354, pp 159–176. Springer, Glasgow, UK
108. Lu Y, Luo Y, Pan A, Mao Y, Xiao J (2023) Domain generalization with global sample mixup. In: Karlinsky L, Michaeli T, Nishino K (eds) Proceedings of the European conference on computer vision—ECCV 2022 Workshops, pp 518–529. Springer, Cham
109. Hendrycks D, Mu N, Cubuk ED, Zoph B, Gilmer J, Lakshminarayanan B (2020) Augmix: a simple data processing method to improve robustness and uncertainty. In: 8th international conference on learning representations. OpenReview.net, Addis Ababa, Ethiopia
110. Yao H, Wang Y, Zhang L, Zou JY, Finn C (2022) C-mixup: improving generalization in regression. *Adv Neural Inf Process Syst* 35:3361–3376

111. Yasuda Y, Onishi R (2022) Spatio-temporal super-resolution data assimilation (SRDA) utilizing deep neural networks with domain generalization technique toward four-dimensional SRDA. *arXiv preprint arXiv:2212.03656*
112. Li X, Dai Y, Ge Y, Liu J, Shan Y, Duan L (2022) Uncertainty modeling for out-of-distribution generalization. In: The tenth international conference on learning representations. OpenReview.net, Virtual Event
113. Li X, Hu Z, Liu J, Ge Y, Dai Y, Duan L-Y (2023) Modeling uncertain feature representation for domain generalization. *CoRR* <https://doi.org/10.48550/arXiv.2301.06442>
114. Zhang Y, Deng B, Li R, Jia K, Zhang L (2023) Adversarial style augmentation for domain generalization. *CoRR* <https://doi.org/10.48550/arXiv.2301.12643>
115. Zhang Z, Wang B, Jha D, Demir U, Bagci U (2022) Domain generalization with correlated style uncertainty. *CoRR* <https://doi.org/10.48550/arXiv.2212.09950>
116. Tan W, Wang P, Ding C, Gong M, Jia K (2022) Style interleaved learning for generalizable person re-identification. *CoRR* <https://doi.org/10.48550/arXiv.2207.03132>
117. Chen L, Zhang Y, Song Y, Wang J, Liu L (2023) Domain generalization via independent regularization from early-branching networks. <https://openreview.net/forum?id=uPPbSJcMBXf>
118. Kim T, Han B (2023) Randomized adversarial style perturbations for domain generalization. *CoRR* <https://doi.org/10.48550/arXiv.2304.01959>
119. Li Q, Huang J, Hu J, Gong S (2022) Feature-distribution perturbation and calibration for generalized person reid. *CoRR* <https://doi.org/10.48550/arXiv.2205.11197>
120. Zhou Q, Zhang K-Y, Yao T, Lu X, Yi R, Ding S, Ma L (2023) Instance-aware domain generalization for face anti-spoofing. *CoRR* <https://doi.org/10.48550/arXiv.2304.05640>
121. Jeon S, Hong K, Lee P, Lee J, Byun H (2021) Feature stylization and domain-aware contrastive learning for domain generalization. In: Proceedings of the 29th ACM international conference on multimedia, pp 22–31. ACM, Virtual Event
122. Wang Y, Qi L, Shi Y, Gao Y (2022) Feature-based style randomization for domain generalization. *IEEE Trans Circuits Syst Video Technol* 32(8):5495–5509. <https://doi.org/10.1109/TCSVT.2022.3152615>
123. Fan Q, Segu M, Tai Y-W, Yu F, Tang C-K, Schiele B, Dai D (2022) Normalization perturbation: a simple domain generalization method for real-world domain shifts. *CoRR* <https://doi.org/10.48550/arXiv.2211.04393>
124. Kang J, Lee S, Kim N, Kwak S (2022) Style neophile: constantly seeking novel styles for domain generalization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 7120–7130. IEEE, New Orleans, LA, USA
125. Zhang L, Liu Z, Zhang W, Zhang D (2023) Style uncertainty based self-paced meta learning for generalizable person re-identification. *IEEE Trans Image Process* 32:2107–2119. <https://doi.org/10.1109/TIP.2023.3263112>
126. Nam H, Lee H, Park J, Yoon W, Yoo D (2021) Reducing domain gap by reducing style bias. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 8690–8699. Computer Vision Foundation/IEEE, virtual
127. Nuriel O, Benaim S, Wolf L (2021) Permuted adain: Reducing the bias towards global statistics in image classification. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 9482–9491. Computer Vision Foundation/IEEE, virtual
128. Tang Z, Gao Y, Zhu Y, Zhang Z, Li M, Metaxas D (2021) Selfnorm and crossnorm for out-of-distribution robustness. *CoRR* [arXiv:2102.02811](https://arxiv.org/abs/2102.02811)
129. Wang Z, Wang Z, Yu Z, Deng W, Li J, Gao T, Wang Z (2022) Domain generalization via shuffled style assembly for face anti-spoofing. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 4113–4123. IEEE, New Orleans, LA, USA
130. Li P, Li D, Li W, Gong S, Fu Y, Hospedales TM (2021) A simple feature augmentation for domain generalization. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 8866–8875. IEEE, Montreal, QC, Canada
131. Wang D, Yang J, Cui W, Xie L, Sun S (2022) Airfi: empowering wifi-based passive human gesture recognition to unseen environment via domain generalization. *IEEE Trans Mobile Comput*
132. Huang Z, Wang H, Xing EP, Huang D (2020) Self-challenging improves cross-domain generalization. In: Proceedings of the European conference on computer vision (ECCV), vol 12347, pp 124–140. Springer, Glasgow, UK
133. Zhao X, Liu C, Sicilia A, Hwang SJ, Fu Y (2022) Test-time Fourier style calibration for domain generalization. In: Raedt LD (ed) Proceedings of the thirty-first international joint conference on artificial intelligence, pp 1721–1727. ijcai.org, Vienna, Austria

134. Hou F, Zhang Y, Liu Y, Yuan J, Zhong C, Zhang Y, Shi Z, Fan J, He Z (2023) Learning how to learn domain-invariant parameters for domain generalization. In: ICASSP 2023–2023 IEEE international conference on acoustics, speech and signal processing (ICASSP), pp 1–5 . IEEE
135. Zheng K, Cao Y, Zhu K, Zhao R, Zha Z-J (2022) Famlp: a frequency-aware mlp-like architecture for domain generalization. CoRR <https://doi.org/10.48550/arXiv.2203.12893>
136. Chen Y, Song P, Liu H, Dai L, Zhang X, Ding R, Li S (2023) Achieving domain generalization for underwater object detection by domain mixup and contrastive learning. *Neurocomputing* 528:20–34. <https://doi.org/10.1016/j.neucom.2023.01.053>
137. Peng D, Pan SJ (2022) Learning gradient-based mixup towards flatter minima for domain generalization. CoRR <https://doi.org/10.48550/arXiv.2209.14742>
138. Lu W, Wang J, Yu H, Huang L, Zhang X, Chen Y, Xie X (2022) Fixed: frustratingly easy domain generalization with mixup. CoRR <https://doi.org/10.48550/arXiv.2211.05228>
139. Shu Y, Cao Z, Wang C, Wang J, Long M (2021) Open domain generalization with domain-augmented meta-learning. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 9624–9633. Computer Vision Foundation / IEEE, virtual
140. Upchurch P, Gardner J, Pleiss G, Pless R, Snaveley N, Bala K, Weinberger K (2017) Deep feature interpolation for image content changes. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 6090–6099. IEEE Computer Society, Honolulu, HI, USA
141. Luo M, Ji S, Wei S (2023) A diverse large-scale building dataset and a novel plug-and-play domain generalization method for building extraction. *IEEE J Sel Top Appl Earth Obs Remote Sens* 16:4122–4138. <https://doi.org/10.1109/JSTARS.2023.3268176>
142. Chang J-R, Wu M-S, Yu W-H, Chen C-C, Yang C-K, Lin Y-Y, Yeh C-Y (2021) Stain mix-up: unsupervised domain generalization for histopathology images. In: Medical image computing and computer assisted intervention–MICCAI 2021: 24th international conference, vol 12903, pp 117–126. Springer, Strasbourg, France
143. Wu L, Huang K, Shen H (2020) A GAN-based tunable image compression system. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 2323–2331. IEEE, Snowmass Village, CO, USA
144. Gupta S, Keshari A, Das S (2022) RV-GAN: Recurrent GAN for unconditional video generation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 2023–2032. IEEE, New Orleans, LA, USA
145. Zhou Y, Liang X, Zhang W, Zhang L, Song X (2021) VAE-based deep SVDD for anomaly detection. *Neurocomputing* 453:131–140. <https://doi.org/10.1016/j.neucom.2021.04.089>
146. Goodfellow I, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, Courville A, Bengio Y (2014) Generative adversarial nets. In: Neural information processing systems, Montreal, Quebec, Canada, pp 2672–2680
147. Kingma DP, Welling M (2014) Auto-encoding variational Bayes. In: 2nd international conference on learning representations. <http://arxiv.org/abs/1312.6114>
148. Yang L, Zhang Z, Song Y, Hong S, Xu R, Zhao Y, Zhang W, Cui B, Yang M (2024) Diffusion models: a comprehensive survey of methods and applications. *ACM Comput Surv* 56(4):105–110539. <https://doi.org/10.1145/3626235>
149. Vaswani A (2017) Attention is all you need. *Adv Neural Inform Process Systems*
150. Zhu J-Y, Park T, Isola P, Efros AA (2017) Unpaired image-to-image translation using cycle-consistent adversarial networks. In: Proceedings of the IEEE international conference on computer vision, pp 2242–2251. IEEE Computer Society, Venice, Italy
151. Anoosheh A, Agustsson E, Timofte R, Van Gool L (2018) Combogan: Unrestrained scalability for image domain translation. In: Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp 783–790. Computer Vision Foundation / IEEE Computer Society, Salt Lake City, UT, USA
152. Choi Y, Choi M, Kim M, Ha J-W, Kim S, Choo J (2018) Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 8789–8797. Computer Vision Foundation / IEEE Computer Society, Salt Lake City, UT, USA
153. Alimisis P, Mademlis I, Radoglou-Grammatikis P, Sarigiannidis P, Papadopoulos GT (2024) Advances in diffusion models for image data augmentation: a review of methods, models, evaluation metrics and future research directions. *arXiv preprint arXiv:2407.04103*
154. Devlin J, Chang M, Lee K, Toutanova K (2019) BERT: pre-training of deep bidirectional transformers for language understanding. In: Burstein J, Doran C, Solorio T (eds) Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language tech-

- nologies, pp 4171–4186. Association for Computational Linguistics, Minneapolis, MN, USA . <https://doi.org/10.18653/V1/N19-1423>
155. Radford A, Wu J, Child R, Luan D, Amodei D, Sutskever I et al (2019) Language models are unsupervised multitask learners. OpenAI blog 1(8):9
 156. Chen X, Hu J, Jin C, Li L, Wang L (2022) Understanding domain randomization for sim-to-real transfer. In: The tenth international conference on learning representations. OpenReview.net, Virtual Event
 157. Pashevich A, Strudel R, Kalevatykh I, Laptev I, Schmid C (2019) Learning to augment synthetic images for sim2real policy transfer. In: 2019 IEEE/RSJ international conference on intelligent robots and systems (IROS), pp 2651–2657. IEEE, Macau, SAR, China
 158. Goodfellow IJ, Shlens J, Szegedy C (2015) Explaining and harnessing adversarial examples. In: Bengio Y, LeCun Y (eds) 3rd international conference on learning representations. San Diego, CA, USA
 159. Sinha A, Namkoong H, Duchi JC (2018) Certifying some distributional robustness with principled adversarial training. In: 6th international conference on learning representations. OpenReview.net, Vancouver, BC, Canada
 160. Peyré G, Cuturi M et al (2019) Computational optimal transport: with applications to data science. *Found Trends Mach Learn* 11(5–6):355–607
 161. Yang Y, Soatto S (2020) Fda: Fourier domain adaptation for semantic segmentation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 4085–4095. Computer Vision Foundation / IEEE, Seattle, WA, USA
 162. Yang Y, Lao D, Sundaramoorthi G, Soatto S (2020) Phase consistent ecological domain adaptation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 9008–9017. Computer Vision Foundation / IEEE, Seattle, WA, USA
 163. Song X, Liu J, Jin Z, Dual Z (2022) Z prototype contrastive learning with Fourier generalization for domain adaptive person re-identification. *Knowl-Based Syst* 256:109851. <https://doi.org/10.1016/j.knsys.2022.109851>
 164. Lu W, Wang J, Li H, Chen Y, Xie X (2022) Domain-invariant feature exploration for domain generalization. *Trans Mach Learn Res* **2022**
 165. Zhang H, Cisse M, Dauphin YN, Lopez-Paz D (2018) Mixup: Beyond empirical risk minimization. In: International conference on learning representations
 166. Pizer SM, Amburn EP, Austin JD, Cromartie R, Geselowitz A, Greer T, Haar Romeny B, Zimmerman JB, Zuiderveld K (1987) Adaptive histogram equalization and its variations. *Comput Vis Graphics Image Process* 39(3):355–368
 167. Wang H, Wu X, Huang Z, Xing EP (2020) High-frequency component helps explain the generalization of convolutional neural networks. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 8681–8691. Computer Vision Foundation / IEEE, Seattle, WA, USA
 168. Yin D, Gontijo Lopes R, Shlens J, Cubuk ED, Gilmer J (2019) A Fourier perspective on model robustness in computer vision. *Adv Neural Inf Process Syst* 32:13255–13265
 169. Camuto A, Willetts M, Simsekli U, Roberts SJ, Holmes CC (2020) Explicit regularisation in gaussian noise injections. *Adv Neural Inf Process Syst* 33:16603–16614
 170. Dumoulin V, Shlens J, Kudlur M (2017) A learned representation for artistic style. In: 5th international conference on learning representations. OpenReview.net, Toulon, France
 171. Huang X, Belongie S (2017) Arbitrary style transfer in real-time with adaptive instance normalization. In: Proceedings of the IEEE international conference on computer vision, pp 1510–1519. IEEE Computer Society, Venice, Italy
 172. Ulyanov D, Vedaldi A, Lempitsky V (2016) Instance normalization: The missing ingredient for fast stylization. *CoRR arXiv:1607.08022*
 173. Sohn K, Berthelot D, Carlini N, Zhang Z, Zhang H, Raffel CA, Cubuk ED, Kurakin A, Li C-L (2020) Fix-match: simplifying semi-supervised learning with consistency and confidence. *Adv Neural Inf Process Syst* 33:596–608
 174. Qi CR, Yi L, Su H, Guibas LJ (2017) Pointnet++: deep hierarchical feature learning on point sets in a metric space. *Adv Neural Inf Process Syst* 30:5099–5108
 175. Xu M, Zhang J, Ni B, Li T, Wang C, Tian Q, Zhang W (2020) Adversarial domain adaptation with domain mixup. In: Proceedings of the AAAI conference on artificial intelligence, vol 34, pp 6502–6509. AAAI Press, New York
 176. Kim D, Seo M, Park K, Shin I, Woo S, Kweon IS, Choi D-G (2023) Bidirectional domain mixup for domain adaptive semantic segmentation. In: Proceedings of the AAAI conference on artificial intelligence, vol 37, pp 1114–1123. AAAI Press, Washington, DC, USA
 177. Luo C, Song C, Zhang Z (2020) Generalizing person re-identification by camera-aware invariance learning and cross-domain mixup. In: Proceedings of the European conference on computer vision (ECCV), vol 12360, pp 224–241. Springer, Glasgow

178. Wu H, Song C, Yue S, Wang Z, Xiao J, Liu Y (2022) Dynamic video mix-up for cross-domain action recognition. *Neurocomputing* 471:358–368. <https://doi.org/10.1016/j.neucom.2021.11.054>
179. Guo H, Mao Y, Zhang, R (2019) Mixup as locally linear out-of-manifold regularization. In: *Proceedings of the AAAI conference on artificial intelligence*, vol 33, pp 3714–3722. AAAI Press, Honolulu, Hawaii, USA
180. Zhang L, Deng Z, Kawaguchi K, Ghorbani A, Zou J (2021) How does mixup help with robustness and generalization? In: *9th international conference on learning representations*. OpenReview.net, Virtual Event, Austria
181. Azizi S, Mustafa B, Ryan F, Beaver Z, Freyberg J, Deaton J, Loh A, Karthikesalingam A, Kornblith S, Chen T, et al. (2021) Big self-supervised models advance medical image classification. In: *Proceedings of the IEEE/CVF international conference on computer vision*, pp 3458–3468. IEEE, Montreal, QC, Canada
182. Zhang X, Jin M, Cheng R, Li R, Han E, Stolcke A (2022) Contrastive-mixup learning for improved speaker verification. *ICASSP 2022–2022 IEEE international conference on acoustics*. Speech and signal processing (ICASSP). IEEE, Virtual and Singapore, pp 7652–7656
183. DeVries T, Taylor GW (2017) Improved regularization of convolutional neural networks with cutout. *CoRR arXiv:1708.04552*
184. Zhang J, Qi L, Shi, Y, Gao Y (2022) Mvdg: a unified multi-view framework for domain generalization. In: *Proceedings of the European conference on computer vision (ECCV)*, vol 13687, pp 161–177. Springer, Tel Aviv, Israel
185. Jing L, Tian Y (2021) Self-supervised visual feature learning with deep neural networks: a survey. *IEEE Trans Pattern Anal Mach Intell* 43(11):4037–4058. <https://doi.org/10.1109/TPAMI.2020.2992393>
186. Li D, Yang Y, Song Y-Z, Hospedales TM (2017) Deeper, broader and artier domain generalization. In: *Proceedings of the IEEE international conference on computer vision*, pp 5543–5551. IEEE Computer Society, Venice, Italy
187. Fang C, Xu Y, Rockmore DN (2013) Unbiased metric learning: On the utilization of multiple datasets and web images for softening bias. In: *Proceedings of the IEEE international conference on computer vision*, pp 1657–1664. IEEE Computer Society, Sydney, Australia
188. Venkateswara H, Eusebio J, Chakraborty S, Panchanathan S (2017) Deep hashing network for unsupervised domain adaptation. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp 5385–5394. IEEE Computer Society, Honolulu, HI, USA
189. LeCun Y, Bottou L, Bengio Y, Haffner P (1998) Gradient-based learning applied to document recognition. *Proc IEEE* 86(11):2278–2324
190. Ganin Y, Lempitsky V (2015) Unsupervised domain adaptation by backpropagation. In: *International conference on machine learning*, vol 37, pp 1180–1189. JMLR.org, Lille, France
191. Netzer Y, Wang T, Coates A, Bissacco A, Wu B, Ng AY (2011) Reading digits in natural images with unsupervised feature learning
192. Zhang H, Song H, Li S, Zhou M, Song D (2024) A survey of controllable text generation using transformer-based pre-trained language models. *ACM Comput Surv* 56(3):64–16437. <https://doi.org/10.1145/3617680>
193. Kaushik D, Setlur A, Hovy EH, Lipton ZC (2021) Explaining the efficacy of counterfactually augmented data. In: *9th International conference on learning representations*. OpenReview.net, Virtual Event, Austria
194. Von Kügelgen J, Sharma Y, Gresele L, Brendel W, Schölkopf B, Besserve M, Locatello F (2021) Self-supervised learning with data augmentations provably isolates content from style. *Adv Neural Inf Process Syst* 34:16451–16467
195. Kamalloo E, Rezagholizadeh M, Passban P, Ghodsi A (2021) Not far away, not so close: sample efficient nearest neighbour data augmentation via minimax. In: *Findings of the association for computational linguistics: ACL/IJCNLP 2021*, vol. ACL/IJCNLP 2021, pp 3522–3533. Association for Computational Linguistics, Online Event
196. Wang H, Lohit S, Jones MN, Fu Y (2022) What makes a "good" data augmentation in knowledge distillation—a statistical perspective. *Adv Neural Inf Process Syst* 35:13456–13469
197. Teney D, Abbasnedjad E, Hengel A (2020) Learning what makes a difference from counterfactual examples and gradient supervision. In: *Proceedings of the European conference on computer vision (ECCV)*, vol 12355, pp 580–599. Springer, Glasgow, UK
198. Chang C-H, Adam GA, Goldenberg (2021) Towards robust classification model by counterfactual and invariant data generation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp 15212–15221. Computer Vision Foundation/IEEE, virtual
199. Wang Z, Culotta A (2021) Robustness to spurious correlations in text classification via automatically generated counterfactuals. In: *Proceedings of the AAAI conference on artificial intelligence*, vol 35, pp 14024–14031. AAAI Press, Virtual Event

200. Cubuk ED, Zoph B, Mane D, Vasudevan V, Le QV (2019) Autoaugment: learning augmentation strategies from data. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 113–123
201. Raileanu R, Goldstein M, Yarats D, Kostrikov I, Fergus R (2021) Automatic data augmentation for generalization in reinforcement learning. *Adv Neural Inf Process Syst* 34:5402–5415
202. Peng X, Tang Z, Yang F, Feris RS, Metaxas D (2018) Jointly optimize data augmentation and network training: adversarial data augmentation in human pose estimation. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 2226–2234
203. Yang X, Che H, Leung M-F, Wen S (2024) Self-paced regularized adaptive multi-view unsupervised feature selection. *Neural Netw* 175:106295
204. Li C, Che H, Leung M-F, Liu C, Yan Z (2023) Robust multi-view non-negative matrix factorization with adaptive graph and diversity constraints. *Inf Sci* 634:587–607
205. Dong Y, Che H, Leung M-F, Liu C, Yan Z (2024) Centric graph regularized log-norm sparse non-negative matrix factorization for multi-view clustering. *Signal Process* 217:109341

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.